

QMC²BO: A Quasi–Monte Carlo Approach to Consensus-Based Optimization and Its Convergence to the Minimizer

S. Göttlich*, J. Heieck*, A. Neuenkirch*

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Abstract

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1 Introduction

A central problem in applied mathematics is the global minimization of a potentially non-convex and non-smooth function $f : \mathbb{R}^D \rightarrow \mathbb{R}$, along with the search for its global minimizer $x^* \in \mathbb{R}^D$.

We study the deterministic particle system $x_t = (x_t^1, \dots, x_t^N) \in \mathbb{R}^{D \cdot N}$, $t \geq 0$, which follows the dynamics

$$\frac{d}{dt}x_t^n = -\lambda(x_t^n - \nu_f^\alpha(x_t)), \quad n = 1, \dots, N, \quad (1.1)$$

with given initial values

$$x_0^n \in [-K, K]^D, \quad n = 1, \dots, N,$$

for some $K > 0$. To simplify our presentation and our analysis, we assume without loss of generality that

$$x_0^n = \mathfrak{s}(\mathbf{p}^n), \quad n = 1, \dots, N,$$

for some given points

$$\mathbf{p}^1, \dots, \mathbf{p}^N \in [0, 1]^D,$$

where the transformation $\mathfrak{s}$ is given by

$$\mathfrak{s} : [0, 1]^D \rightarrow [-K, K]^D, \quad \mathfrak{s}(u) = 2Ku - K\mathbf{1}_D. \quad (1.2)$$

The consensus function $\nu_f^\alpha : \mathbb{R}^{D \times N} \rightarrow \mathbb{R}^D$ is given by

$$\nu_f^\alpha(x_t) = \sum_{m=1}^N x_t^m \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))}. \quad (1.3)$$

This system arises from the classical CBO particle systems by setting $\sigma = 0$ and will be called zero-noise CBO particle system.

The system (1.1) is a classical flocking model and admits a unique solution if $f : \mathbb{R}^D \rightarrow \mathbb{R}$ is locally Lipschitz. Moreover, it is also well known that a consensus x_∞ emerges for $t \rightarrow \infty$ which belongs to the convex hull of the initial points, that is, we have

$$\exists x_\infty \in \text{Conv}(x_0^1, \dots, x_0^N) : \lim_{t \rightarrow \infty} x_t^n = x_\infty, \quad n = 1, \dots, N.$$

The crucial question now is how far apart x_∞ and x^* are. Note that the consensus x_∞ depends on the initial points $x_0^1, \dots, x_0^N$ and on λ and α , while the global minimizer x^* is independent of these quantities.

¹University of Mannheim, Department of Mathematics, 68131 Mannheim, Germany (goettlich@math.uni-mannheim.de, jacob.heieck@uni-mannheim.de, neuenkirch@uni-mannheim.de).

In the next chapter, we derive a precise description of the convergence to consensus by establishing the global exponential stability of the system (1.1). This analysis reveals the rate at which the particles contract towards the limiting consensus point x_∞ and clarifies the underlying geometric structure of the flow. In Chapter 3 we will then use this convergence behavior to rigorously estimate the deviation between the average particle position $\bar{x}_t^N$ and the global minimizer x^* . In particular, we quantify how this discrepancy depends on the number of particles N and the concentration parameter α . Chapter 4 presents numerical experiments that illustrate the theoretical findings. Finally, Chapters 5 and ?? situate the method within the broader literature and provide concluding remarks.

2 Global Exponential Stability of the QMC²BO Dynamics

The purpose of this chapter is to provide a rigorous analysis of the consensus formation behavior of the QMC²BO dynamics (1.1). Our goal is to establish *global exponential stability*, which in particular yields a quantitative description of how fast the system approaches its limiting consensus state x_∞ . To this end, we begin by rewriting the dynamics in a form that highlights the geometric structure driving the consensus formation.

This reformulation naturally leads to the introduction of classical notions from Lyapunov stability theory, well suited for analyzing multi-particle systems. Within this framework, we show that the dynamics converge exponentially towards a consensus subspace, with the convergence rate given by λ .

2.1 Reformulation of the QMC²BO System

By definition of the consensus function $\nu_f^\alpha(x_t)$ in (1.3), the QMC²BO dynamics (1.1) is given by

$$\frac{dx_t^n}{dt} = -\lambda \left(x_t^n - \sum_{m=1}^N a_m(x_t) x_t^m \right), \quad n = 1, \dots, N,$$

with weights

$$a_m = a_m(x_t) = \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))}.$$

Note that $a_m = a_m(x_t)$ depends non-linearly on all particle positions x_t . In the following, we will often drop this dependence for notational easiness.

Since

$$\sum_{m=1}^N a_m = \sum_{m=1}^N \frac{\exp(-\alpha f(x_t^m))}{\sum_{k=1}^N \exp(-\alpha f(x_t^k))} = 1,$$

all weights sum up to 1 and we can rewrite

$$\frac{dx_t^n}{dt} = -\lambda \left(\sum_{\substack{m=1 \\ m \neq n}}^N a_m \cdot (x_t^n - x_t^m) \right).$$

Taking all N equations simultaneously into account and sorting them by spatial dimension d yields

$$dx_t = d \begin{pmatrix} x_t^{1,1} \\ \vdots \\ x_t^{N,1} \\ \vdots \\ x_t^{1,D} \\ \vdots \\ x_t^{N,D} \end{pmatrix} = -\lambda \begin{pmatrix} \sum_{m=2}^N a_m \cdot (x_t^{1,1} - x_t^{m,1}) \\ \vdots \\ \sum_{m=1}^{N-1} a_m \cdot (x_t^{N,1} - x_t^{m,1}) \\ \vdots \\ \sum_{m=2}^N a_m \cdot (x_t^{1,D} - x_t^{m,D}) \\ \vdots \\ \sum_{m=1}^{N-1} a_m \cdot (x_t^{N,D} - x_t^{m,D}) \end{pmatrix} dt. \quad (2.1)$$

Given the weights $a_1, \dots, a_N$, the evolution of each dimension is therefore independent of the others, as it depends only on the interactions between particles within the same dimension. Moreover, the weights a_m are equal in all equations.

Thus, it will suffice to study the stability of the system in each dimension $d \in \{1, \dots, D\}$ on its own. If we consider the system

$$d \begin{pmatrix} x_t^{1,d} \\ \vdots \\ x_t^{N,d} \end{pmatrix} = -\lambda \begin{pmatrix} \sum_{m=2}^N a_m \cdot (x_t^{1,d} - x_t^{m,d}) \\ \vdots \\ \sum_{m=1}^{N-1} a_m \cdot (x_t^{N,d} - x_t^{m,d}) \end{pmatrix} dt \quad (2.2)$$

and introduce the notation $x_t^{\cdot,d} = (x_t^{1,d}, \dots, x_t^{N,d})^\top \in \mathbb{R}^N$, the right-hand side of the corresponding system simplifies to

$$-\lambda \hat{L}(x_t) x_t^{\cdot,d} dt,$$

where $\hat{L}(x_t)$ is defined as follows

$$\hat{L}(x_t) := \begin{pmatrix} 1 - a_1(x_t) & -a_2(x_t) & \cdots & -a_N(x_t) \\ -a_1(x_t) & 1 - a_2(x_t) & \cdots & -a_N(x_t) \\ \vdots & \vdots & \ddots & \vdots \\ -a_1(x_t) & -a_2(x_t) & \cdots & 1 - a_N(x_t) \end{pmatrix}. \quad (2.3)$$

The diagonal entries arise since the weights sum up to 1. This leads to the compact representation

$$dx_t = -\lambda L(x_t) x_t dt \quad (2.4)$$

of system (2.1) with

$$L(x_t) = \begin{pmatrix} \hat{L}(x_t) & \mathbf{0}_{N \times N} & \cdots & \mathbf{0}_{N \times N} \\ \mathbf{0}_{N \times N} & \hat{L}(x_t) & \cdots & \mathbf{0}_{N \times N} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{N \times N} & \mathbf{0}_{N \times N} & \cdots & \hat{L}(x_t) \end{pmatrix} \in \mathbb{R}^{D \cdot N \times D \cdot N}. \quad (2.5)$$

Here, $\mathbf{0}_{N \times N}$ denotes the zero-matrix of dimension $N \times N$. This formulation emphasizes that each dimension d can be treated separately, while still preserving the component-wise non-linear interactions between particles through the weights $a_1, \dots, a_N$.

From a graph-theoretic perspective, the matrix $\hat{L}(x_t)$ can be seen as the Laplacian [?] of a weighted complete graph, where the nodes represent the particles $x_t^{n,d}$ and the edges carry the interaction weights $-a_m(x_t)$ or $1 - a_n(x_t)$. Each off-diagonal $-a_m(x_t)$ entry describes how strongly the particle n is influenced by the particle m , while the diagonal terms ensure that the total influence on each particle is balanced. The motion of the particle n is exactly a weighted combination of the positions of all other particles, with higher weights meaning that n is pulled more strongly toward those particular particles.

This framework links to the classical theory of multi-agent consensus, where stability has been fully characterized in the linear case. In contrast, the QMC²BO system involves highly nonlinear weights $a_m(x_t)$, which require a refined analysis. The next section develops stability results for this nonlinear setting. In the following, we will drop the dependence of $\hat{L} = \hat{L}(x_t)$ and $L = L(x_t)$ on x_t for notational simplicity.

2.2 Lyapunov-Based Stability Analysis

Having reformulated the system dynamics, we now turn to the analysis of its stability properties. We aim to show that the component-wise average of the system

$$\bar{x}_t^N = \begin{pmatrix} \bar{x}_t^{N,1} \\ \vdots \\ \bar{x}_t^{N,D} \end{pmatrix} = \begin{pmatrix} \frac{1}{N} \sum_{k=1}^N x_t^{k,1} \\ \vdots \\ \frac{1}{N} \sum_{k=1}^N x_t^{k,D} \end{pmatrix}, \quad t \geq 0, \quad (2.6)$$

is exponentially attracting for the particle system. More precisely, we will show that

$$x_t^{n,d} = \bar{x}_t^{N,d} + e^{-\lambda t} (x_0^{n,d} - \bar{x}_0^{N,d}), \quad t \geq 0, \quad (2.7)$$

for all $d = 1, \dots, D$ and $n = 1, \dots, N$.

We proceed by precisely defining the relevant stability concept for our analysis. Intuitively, stability describes the persistence of solutions under small perturbations. In particular, *global stability* means that this property holds for any initial condition $x_0 \in \mathbb{R}^N$, not just for states near a solution. *Exponential stability* further quantifies the rate of convergence, requiring that deviations from the solution decay at least exponentially fast over time.

We now formalize these notions.

Definition 2.1 (Exponential Stability for ODE). The point $\mathbf{0}_N$ is called *globally exponentially stable* if there exist constants $c > 0$ and $\delta > 0$ such that

$$\|x_t\| \leq c \|x_0\| e^{-\delta t}, \quad t \geq 0,$$

for any $x_0 \in \mathbb{R}^N$.

Notice that Definition 2.1 applies specifically to the point $\mathbf{0}_N$. The stability of other equilibrium points or entire sets of points can be analyzed through a suitable transformation of the state space.

To assess global exponential stability, one may employ a fundamental mathematical tool: Lyapunov functions. These functions provide a systematic approach to determining whether deviations from the solution of the dynamical system decay exponentially fast.

Definition 2.2 (Lyapunov functions for ODE). Let $V : \mathbb{R}^N \rightarrow \mathbb{R}$ be a positive definite function with continuous partial derivatives. If along every trajectory of the system $dx_t = a(x_t)dt$ its time derivative satisfies

$$\dot{V}(x_t) := \frac{dV(x_t)}{dt} = (\nabla V(x_t))^\top \frac{dx_t}{dt} = (\nabla V(x_t))^\top a(x_t) \leq 0$$

for all $t \geq 0$, then V is called a *Lyapunov function* for the system.

The equalities in Definition 2.2 follow directly from the chain rule, using that V depends only on the state x_t and not explicitly on time t . If V satisfies additional regularity and growth conditions, one can further conclude that a solution is globally exponentially stable.

Theorem 2.3 (Exponential Stability for ODE). Consider the ordinary differential equation $dx_t = a(x_t)dt$ for $t \geq 0$ and assume that $\mathbf{0}_N$ is an equilibrium, i.e. $a(\mathbf{0}_N) = \mathbf{0}_N$. Let $V : \mathbb{R}^N \rightarrow \mathbb{R}$ be a Lyapunov function for this system, and suppose that there exist constants $c_1, c_2, c_3, c_4 > 0$ such that

- (i) $c_1 \|x\|^2 \leq V(x) \leq c_2 \|x\|^2$ for $x \in \mathbb{R}^N$,
- (ii) $\dot{V}(x_t) \leq -c_3 \|x_t\|^2$ for all $t \geq 0$,
- (iii) $\|\nabla V(x)\| \leq c_4 \|x\|$ for $x \in \mathbb{R}^N$.

Then the equilibrium $\mathbf{0}_N$ is globally exponentially stable, with

$$c = \sqrt{\frac{c_2}{c_1}}, \quad \delta = \frac{c_3}{2c_2}.$$

The equilibrium points of the QMC²BO system (1.1) are the solutions of

$$-\lambda (x_t^n - \nu_f^\alpha(x_t)) = \mathbf{0}_N \quad \Leftrightarrow \quad x_t^n = \nu_f^\alpha(x_t)$$

for all $n = 1, \dots, N$. As the right-hand side depends on the full set of particle positions and not on individual entries, a necessary condition for equilibrium is

$$x_t^1 = \dots = x_t^N. \tag{2.8}$$

To verify whether this condition is also sufficient, assume that (2.8) holds. As $\nu_f^\alpha(X_t)$ is a weighted average of all particle positions, we compute for any $n \in \{1, \dots, N\}$:

$$x_t^n = \sum_{m=1}^N \frac{x_t^m}{N} = \sum_{m=1}^N \frac{x_t^m \exp(-\alpha f(X_t^m))}{\sum_{k=1}^N \exp(-\alpha f(X_t^k))} = \nu_f^\alpha(x_t),$$

which confirms the necessary condition.

Thus, the set of equilibrium points consists precisely of those states in which all particles agree component-wise. In particular, the equilibrium set is

$$\mathcal{G}^* = \{x \in \mathbb{R}^{N \cdot D} : x^{1,d} = \dots = x^{N,d}, d = 1, \dots, D\}. \quad (2.9)$$

We have established that the QMC²BO system (1.1) is equivalent to (2.4) and that the dynamics decouple across dimensions $d \in \{1, \dots, D\}$. Hence, we study the one-dimensional subsystem

$$dx_t^{\cdot d} = -\lambda \hat{L} x_t^{\cdot d} dt \quad (2.10)$$

where the matrix $\hat{L}$ is defined in (2.3).

To analyze the stability of this subsystem, we need to understand

$$\dot{V}(x_t^{\cdot d}) = (\nabla V(x_t^{\cdot d}))^\top a(x_t^{\cdot d}) = -\lambda (\nabla V(x_t^{\cdot d}))^\top \hat{L} x_t^{\cdot d}, \quad (2.11)$$

for a suitable function $V : \mathbb{R}^N \rightarrow [0, \infty)$. Establishing stability requires showing that $\dot{V}(x_t^{\cdot d}) \leq 0$, which in turn depends on how the matrix $\hat{L}$ acts on the state vector $x_t^{\cdot d}$. This motivates a closer look at the spectral properties of $\hat{L}$. Its eigenvalues and eigenvectors determine precisely in which directions the dynamics contract and in which directions motion persists.

In order to compute the eigenvalues of $\hat{L}$ explicitly, we make use of the following result on rank-one matrix updates, which allows for efficient determinant calculations.

Lemma 2.4 (Rank-One Update). If $A \in \mathbb{R}^{N \times N}$ is non-singular, and if c, d are N -dimensional column vectors, then:

- (i) $\det(I_N + cd^\top) = 1 + d^\top c$,
- (ii) $\det(A + cd^\top) = \det(A) (1 + d^\top A^{-1}c)$.

We now apply this lemma to derive the eigenvalues of $\hat{L}$. Recall that $\mathbf{1}_N$ denotes the all-ones column vector, and define the weight vector

$$\ell := (a_1, \dots, a_N)^\top \in (0, 1)^N, \quad (2.12)$$

which collects the weights associated with each particle x_t^n , $n = 1, \dots, N$.

Lemma 2.5 (Eigenvalues of $\hat{L}$). The matrix $\hat{L} \in \mathbb{R}^{N \times N}$ has exactly one eigenvalue equal to zero, while the remaining $N - 1$ eigenvalues are equal to 1, that is,

$$\mu_1(\hat{L}) = 0, \quad \mu_2(\hat{L}) = \dots = \mu_N(\hat{L}) = 1.$$

A right-eigenvector corresponding to $\mu_1(\hat{L})$ is $\mathbf{1}_N$, a left-eigenvector is ℓ .

Proof. We begin by rewriting $\hat{L}$ using a rank-one update involving $\mathbf{1}_N$ and $\ell = (a_1, \dots, a_N)^\top$:

$$\hat{L} = \begin{pmatrix} 1 - a_1 & -a_2 & \cdots & -a_N \\ -a_1 & 1 - a_2 & \cdots & -a_N \\ \vdots & \vdots & \ddots & \vdots \\ -a_1 & -a_2 & \cdots & 1 - a_N \end{pmatrix} = I_N - \begin{pmatrix} a_1 & a_2 & \cdots & a_N \\ a_1 & a_2 & \cdots & a_N \\ \vdots & \vdots & \ddots & \vdots \\ a_1 & a_2 & \cdots & a_N \end{pmatrix} = I_N - \mathbf{1}_N \ell^\top.$$

To determine the eigenvalues, we compute the characteristic equation by evaluating $\det(\hat{L} - \mu I_N) = 0$. Using the rank-one update representation of $\hat{L}$ and applying (ii) of Lemma 2.4 with $A = (1 - \mu)I_N$, $c = -\mathbf{1}_N$ and $d = \ell$, we obtain

$$\det(\hat{L} - \mu I_N) = \det((1 - \mu)I_N) (1 - \ell^\top ((1 - \mu)I_N)^{-1} \mathbf{1}_N) = (1 - \mu)^N \left(1 - \frac{1}{1 - \mu} \ell^\top \mathbf{1}_N\right).$$

Here, we use $A^{-1} = \frac{1}{1 - \mu} I_N$ and $\det(A) = (1 - \mu)^N$. Since all weights sum to one, we have

$$\ell^\top \mathbf{1}_N = \sum_{n=1}^N a_n = 1.$$

Substituting this into the equation gives

$$\det(\hat{L} - \mu I_N) = (1 - \mu)^N \left(1 - \frac{1}{1 - \mu}\right).$$

As a result, the determinant satisfies $\det(\hat{L} - \mu I_N) = 0$ if either

$$(1 - \mu)^N = 0 \quad \text{or} \quad 1 - \frac{1}{1 - \mu} = 0.$$

The second condition simplifies to $1 - \mu = 1$, which holds for $\mu = 0$. The first condition implies $\mu = 1$. Thus, the eigenvalues of $\hat{L}$ are

$$\mu_1(\hat{L}) = 0, \quad \mu_2(\hat{L}) = \dots = \mu_N(\hat{L}) = 1.$$

For the corresponding eigenvectors to $\mu_1(\hat{L}) = 0$ we compute

$$\hat{L}\mathbf{1}_N = (I_N - \mathbf{1}_N \ell^\top) \mathbf{1}_N = \mathbf{1}_N - \mathbf{1}_N \ell^\top \mathbf{1}_N = \mathbf{1}_N - \mathbf{1}_N = \mathbf{0}_N = 0 \cdot \mathbf{1}_N = \mu_1(\hat{L}) \cdot \mathbf{1}_N$$

for the right-eigenvector and

$$\hat{L}^\top \ell = (I_N - \mathbf{1}_N \ell^\top)^\top \ell = \ell - \ell \mathbf{1}_N^\top \ell = \ell - \ell = \mathbf{0}_N = 0 \cdot \ell = \mu_1(\hat{L}) \cdot \ell$$

for the left-eigenvector. Thus, we conclude that both $\mathbf{1}_N$ and ℓ are right- and left-eigenvectors corresponding to the eigenvalue $\mu_1(\hat{L}) = 0$. The key property used in this derivation is that the sum of all weights equals one. $\square$

Lemma 2.5 reveals that $\hat{L}$ has exactly one eigenvalue equal to zero, while all remaining $N-1$ eigenvalues are equal to one. The right-eigenvector associated with the zero eigenvalue is $\mathbf{1}_N$, the all-ones vector.

By restricting condition (2.8) to a fixed dimension d , we find that all equilibrium points of the QMC²BO system (1.1) satisfy

$$x_t^{1,d} = \dots = x_t^{N,d}, \quad (2.13)$$

i.e., all particle components agree. Thus, any equilibrium point lies in the one-dimensional subspace spanned by the eigenvector $\mathbf{1}_N$. Thus, the eigenspace associated with the zero eigenvalue corresponds to a direction in which the system does not evolve. Once all particles are aligned, the dynamics stops changing. In contrast, all directions orthogonal to $\mathbf{1}_N$ correspond to the eigenvalue one and are actively contracted by the dynamics. As a result, the system is continuously pushed toward the subspace

$$\mathcal{G}^{\star,d} = \{x \in \mathbb{R}^N : x = \gamma \mathbf{1}_N, \gamma \in \mathbb{R}\}.$$

Motivated by these observations, we consider the function

$$V(x_t^{\cdot,d}) = \|x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N\|^2, \quad (2.14)$$

where $\bar{x}_t^{N,d}$ denotes the average particle value in dimension d , defined in (2.6). Since $\bar{x}_t^{N,d}$ is the orthogonal projection of $x_t^{\cdot,d}$ onto $\mathcal{G}^{\star,d}$, we also have

$$V(x_t^{\cdot,d}) = \text{dist}(x_t^{\cdot,d}; \mathcal{G}^{\star,d})^2 := \inf_{y \in \mathcal{G}^{\star,d}} \|x_t^{\cdot,d} - y\|^2.$$

Thus, V measures the squared distance of the particle system to its closest equilibrium point. Clearly, the difference $x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N \in \mathbb{R}^N$ captures how far each individual particle deviates from the mean, so an alternative interpretation of V is that this function measures the squared distance of $x_t^{\cdot,d}$ to its mean value. However, note that V is not a classical Lyapunov function, since that we have $V(x) = 0$ for all $x \in \mathcal{G}^{\star,d}$.

Now define the auxiliary functions

$$g : \mathbb{R}^N \rightarrow \mathbb{R}, \quad g(x) = x^\top x, \quad h : \mathbb{R}^N \rightarrow \mathbb{R}^N, \quad h(x) = x - \frac{1}{N} \mathbf{1}_N^\top x \mathbf{1}_N,$$

which satisfy

$$\nabla g(x) = 2x, \quad Jh(x) = I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top$$

for $x \in \mathbb{R}^N$. The expression for the Jacobi-matrix of h follows from the observation that

$$\partial_{x_m}(h(x))_k = \partial_{x_m} \left(x_k - \frac{1}{N} \mathbf{1}_N^\top x \right) = \mathbb{1}_{\{m=k\}} - \frac{1}{N}$$

for $m, k = 1, \dots, N$. Here, $\mathbb{1}_{\{m=k\}}$ equals 1 when $m = k$ and 0 otherwise.

Thus, we have

$$V(x) = g(h(x))$$

and so the chain rule yields

$$\begin{aligned} \nabla V(x) &= \left(I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top \right)^\top 2 \left(x - \frac{1}{N} \mathbf{1}_N^\top x \mathbf{1}_N \right) \\ &= 2 \left(x - \frac{1}{N} \mathbf{1}_N^\top x \mathbf{1}_N \right) - \frac{2}{N} \mathbf{1}_N \mathbf{1}_N^\top x + \frac{2}{N^2} \mathbf{1}_N \mathbf{1}_N^\top \mathbf{1}_N \mathbf{1}_N^\top x \\ &= 2 \left(x - \frac{1}{N} \mathbf{1}_N^\top x \mathbf{1}_N \right). \end{aligned}$$

Here, we have used that $x^\top \mathbf{1}_N = \mathbf{1}_N^\top x$ is a scalar and $\mathbf{1}_N^\top \mathbf{1}_N = N$.

Using this expression for ∇V in (2.11) we obtain

$$\dot{V}(x_t^{\cdot,d}) = -2\lambda(x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N)^\top \hat{L} x_t^{\cdot,d} \quad (2.15)$$

Now, we will rewrite and simplify the expression on the right-hand side of the above equality. To this end, we use the representation $\hat{L} = I_N - \mathbf{1}_N \ell^\top$ established in the proof of Lemma 2.5

$$\hat{L} \left(\bar{x}_t^{N,d} \mathbf{1}_N \right) = (I_N - \mathbf{1}_N \ell^\top) \left(\frac{1}{N} \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N \right) = \frac{1}{N} \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N - \frac{1}{N} \mathbf{1}_N \ell^\top \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N.$$

Since $\mathbf{1}_N^\top x_t^{\cdot,d}$ is a scalar, we can rearrange terms and obtain

$$\hat{L} \left(\bar{x}_t^{N,d} \mathbf{1}_N \right) = \frac{1}{N} \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N - \frac{1}{N} \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N \cdot \ell^\top \mathbf{1}_N = 0, \quad (2.16)$$

where we used that the weights sum to one,

$$\ell^\top \mathbf{1}_N = \sum_{n=1}^N a_n = 1.$$

As a result, we can rewrite the total derivative as

$$\dot{V}(x_t^{\cdot,d}) = -2\lambda(x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N)^\top \hat{L} x_t^{\cdot,d} = -2\lambda(x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N)^\top \hat{L} (x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N),$$

since the additional term vanishes due to (2.16). For brevity, we define the residual vector of the d -th component

$$r_t^{\cdot,d} := x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N \quad (2.17)$$

and compute

$$\begin{aligned} \dot{V}(x_t^{\cdot,d}) &= -2\lambda(r_t^{\cdot,d})^\top \hat{L} r_t^{\cdot,d} \\ &= -2\lambda(r_t^{\cdot,d})^\top (I_N - \mathbf{1}_N \ell^\top) r_t^{\cdot,d} \\ &= -2\lambda(r_t^{\cdot,d})^\top r_t^{\cdot,d} + 2\lambda(r_t^{\cdot,d})^\top \mathbf{1}_N \ell^\top r_t^{\cdot,d}. \end{aligned} \quad (2.18)$$

To simplify the second term, we substitute back $r_t^{\cdot,d} = x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N$, use that $(x_t^{\cdot,d})^\top \mathbf{1}_N$ is a scalar and $\mathbf{1}_N^\top \mathbf{1}_N = N$:

$$\begin{aligned} (r_t^{\cdot,d})^\top \mathbf{1}_N \ell^\top &= \left(x_t^{\cdot,d} - \frac{1}{N} \mathbf{1}_N^\top x_t^{\cdot,d} \mathbf{1}_N \right)^\top \mathbf{1}_N \ell^\top \\ &= (x_t^{\cdot,d})^\top \mathbf{1}_N \ell^\top - \frac{1}{N} \mathbf{1}_N^\top (x_t^{\cdot,d})^\top \mathbf{1}_N \mathbf{1}_N \ell^\top \\ &= (x_t^{\cdot,d})^\top \mathbf{1}_N \ell^\top - \frac{N}{N} (x_t^{\cdot,d})^\top \mathbf{1}_N \ell^\top \\ &= \mathbf{0}_N^\top. \end{aligned}$$

Hence, the second term in (2.18) vanishes, and observing that $V(x_t^{\cdot,d}) = \|r_t^{\cdot,d}\|^2$ we obtain

$$\dot{V}(x_t^{\cdot,d}) = -2\lambda(r_t^{\cdot,d})^\top r_t^{\cdot,d} = -2\lambda\|r_t^{\cdot,d}\|^2 = -2\lambda V(x_t^{\cdot,d}).$$

Solving this linear ODE yields

$$V(x_t^{\cdot,d}) = e^{-2\lambda t} V(x_0^{\cdot,d}), \quad t \geq 0, \quad (2.19)$$

so this distance of the particle system to its subspace of equilibrium points decays exponentially with rate λ .

At this point, we require a change of coordinates, if we want to work in the setup of Theorem 2.3, since we have a whole subspace $\mathcal{G}^{*,d}$ of equilibrium point instead of $\mathbf{0}_N$. This is also reflected by the fact that V can not be a Lyapunov function, since $V(x) = 0$ for all $x \in \mathcal{G}^{*,d}$.

From equation (2.19) we already know that

$$\|r_t^{\cdot,d}\|^2 = e^{-2\lambda t} \|r_0^{\cdot,d}\|^2, \quad t \geq 0, \quad (2.20)$$

so we will now turn to $r_t^{\cdot,d} = x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N$, which is the dynamics of the particle system that is orthogonal to $\mathcal{G}^{*,d}$. In fact, we have

$$r_t^{\cdot,d} = x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N = x_t^{\cdot,d} - \frac{1}{N} \mathbf{1}_N^T x_t^{\cdot,d} \mathbf{1}_N = \left(I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^T \right) x_t^{\cdot,d}$$

and so

$$\frac{dr_t^{\cdot,d}}{dt} = P \frac{dx_t^{\cdot,d}}{dt} = -\lambda P \hat{L} x_t^{\cdot,d},$$

where

$$P = I_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^T$$

is the matrix which corresponds to the projection on the orthogonal complement of $\mathcal{G}^{*,d}$. Since

$$P \mathbf{1}_N = \mathbf{1}_N - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^T \mathbf{1}_N = \mathbf{0}_N,$$

we have

$$P \hat{L} = P - P \mathbf{1}_N \ell^T = P. \quad (2.21)$$

Thus, the dynamics of $r_t^{\cdot,d}$ is given by the linear ODE

$$\dot{r}_t^{\cdot,d} = -\lambda r_t^{\cdot,d},$$

which strikingly clarifies the dynamics of the particle system. This ODE has the unique solution

$$r_t^{\cdot,d} = e^{-\lambda t} r_0^{\cdot,d}, \quad t \geq 0, \quad (2.22)$$

which is obviously exponentially stable with rate λ at the unique fixed point $\mathbf{0}_N$ and we trivially recover equation (2.20).

Since equation (2.22) does not depend on the considered dimension d of the particles, we have obtained the following result for the original dynamics:

Theorem 2.6 (Global Exponential Stability of QMC²BO). Consider the QMC²BO system

$$\frac{dx_t^n}{dt} = -\lambda (x_t^n - \nu_f^\alpha(x_t)), \quad n = 1, \dots, N,$$

with $x_t = (x_t^1, \dots, x_t^N)^\top = (x_t^{1,1}, \dots, x_t^{1,D}, \dots, x_t^{N,1}, \dots, x_t^{N,D})^\top \in \mathbb{R}^{D \cdot N}$. Then the system is *globally exponentially stable* with exact rate λ with respect to the set of equilibrium points

$$\mathcal{G}^* = \{x \in \mathbb{R}^{N \cdot D} : x^{1,d} = \dots = x^{N,d}, d = 1, \dots, D\}.$$

More precisely, we have the exact exponential decay

$$\|x_t^{\cdot,d} - \bar{x}_t^{N,d} \mathbf{1}_N\| = e^{-\lambda t} \|x_0^{n,d} - \bar{x}_0^{N,d} \mathbf{1}_N\|, \quad t \geq 0.$$

Moreover, the evolution of each particle can be expressed explicitly as

$$x_t^{n,d} = \bar{x}_t^{N,d} + e^{-\lambda t} (x_0^{n,d} - \bar{x}_0^{N,d}), \quad n = 1, \dots, N, d = 1, \dots, D,$$

where $\bar{x}_t^{N,d}$ denotes the average of the d -th component across all particles.

2.3 Numerical Comparison with Classical CBO and Motivation of QMC²BO

Here, we want to compare the numerical results from the exponential stability analysis of QMC²BO to the one of the classical CBO. It shall motivate, why for general observations, it is useful to omit the stochastic part.

3 Convergence of the QMC²BO Method to the Minimizer

Our next goal is to quantify how close the consensus point x_∞ , to which each particle converges exponentially fast, lies to the true minimizer x^* . In particular, we aim to understand how this discrepancy depends on the concentration parameter α and on the number of particles N . To make this dependence explicit, we begin by rewriting the system in a form better suited for analysis.

From Theorem 2.6 we know that the component-wise average $\bar{x}_t^N$ is exponentially attracting for the particle system. More precisely, for each dimension $d \in \{1, \dots, D\}$ and every particle $n \in \{1, \dots, N\}$,

$$x_t^{n,d} = \bar{x}_t^{N,d} + e^{-\lambda t}(x_0^{n,d} - \bar{x}_0^{N,d}), \quad t \geq 0.$$

Thus, deviations of individual particles from the mean decay exponentially. On the other hand, taking the average of the particle system (1.1) yields the evolution equation

$$\frac{d}{dt} \bar{x}_t^N = \frac{1}{N} \sum_{n=1}^N \frac{d}{dt} x_t^n = \frac{1}{N} \sum_{n=1}^N -\lambda (x_t^n - \nu_f^\alpha(x_t)) = -\lambda (\bar{x}_t^N - \nu_f^\alpha(x_t)).$$

To separate the mean behavior from deviations, define the residual vector

$$r_t = (r_t^1, \dots, r_t^N)^\top = (r_t^{1,1}, \dots, r_t^{1,D}, \dots, r_t^{N,1}, \dots, r_t^{N,D})^\top, \quad (3.1)$$

where $r_t^{n,d} := x_t^{n,d} - \bar{x}_t^{N,d}$ measures the deviation of particle n in dimension d from the particle average. Theorem 2.6 then implies the compact representation

$$x_t = \begin{pmatrix} \bar{x}_t^N \\ \vdots \\ \bar{x}_t^N \end{pmatrix} + e^{-\lambda t} r_0, \quad (3.2)$$

Substituting this into the mean dynamics yields

$$\frac{d}{dt} \bar{x}_t^N = -\lambda (\bar{x}_t^N - \nu_f^\alpha((\bar{x}_t^N, \dots, \bar{x}_t^N)^\top + e^{-\lambda t} r_0)). \quad (3.3)$$

This shows that the evolution of the D -dimensional average is driven by a perturbed consensus function

A closer inspection of the consensus reveals that

$$\begin{aligned} \nu_f^\alpha((\bar{x}_t^N, \dots, \bar{x}_t^N)^\top + e^{-\lambda t} r_0) &= \sum_{m=1}^N (\bar{x}_t^N + e^{-\lambda t} r_0^m) \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))} \\ &= \bar{x}_t^N + e^{-\lambda t} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))} \end{aligned}$$

for $t \geq 0$, using the fact that the weights sum to one. Substituting this into (3.3) yields

$$\frac{d}{dt} \bar{x}_t^N = \lambda e^{-\lambda t} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(\bar{x}_t^N + e^{-\lambda t} r_0^k))}.$$

We now isolate how the particle average $\bar{x}_t^N$ approaches to the minimizer x^* . Substituting back (3.2) yields

$$\begin{aligned} \frac{d}{dt} \bar{x}_t^N &= \lambda \sum_{m=1}^N (x_t^{m,N} - \bar{x}_t^N) \frac{\exp(-\alpha f(x_t^{m,N}))}{\sum_{k=1}^N \exp(-\alpha f(x_t^{k,N}))} \\ &= \lambda (x^* - \bar{x}_t^N) + \lambda \sum_{m=1}^N (x_t^{m,N} - x^*) \frac{\exp(-\alpha f(x_t^{m,N}))}{\sum_{k=1}^N \exp(-\alpha f(x_t^{k,N}))}, \end{aligned}$$

where we added and subtracted $x^\star$ and used again that the weights sum to one.

Define now

$$F_{N,\alpha}(\bar{x}_t^N, t) := \sum_{m=1}^N (x_t^{m,N} - x^\star) \frac{\exp(-\alpha f(x_t^{m,N}))}{\sum_{k=1}^N \exp(-\alpha f(x_t^{k,N}))} \quad (3.4)$$

and introduce the deviation of the mean from the minimizer,

$$\xi_t := x^\star - \bar{x}_t^N. \quad (3.5)$$

Then

$$x_t^{m,N} - e^{-\lambda t} r_0^m = \bar{x}_t^N = x^\star - \xi_t,$$

and

$$\frac{d}{dt} \bar{x}_t^N = \lambda(x^\star - \bar{x}_t^N) + F_{N,\alpha}(\bar{x}_t^N, t) = \lambda \xi_t + \lambda F_{N,\alpha}(x^\star - \xi_t, t).$$

The dynamics of ξ_t satisfy

$$\frac{d}{dt} \xi_t = -\frac{d}{dt} \bar{x}_t^N = -\lambda \xi_t - \lambda F_{N,\alpha}(x^\star - \xi_t, t).$$

Multiplying by $e^{\lambda t}$ shows that

$$e^{\lambda t} \frac{d}{dt} \xi_t + \lambda e^{\lambda t} \xi_t = -e^{\lambda t} \lambda F_{N,\alpha}(x^\star - \xi_t, t).$$

The left-hand side coincides with the time derivative of the product $e^{\lambda t} \xi_t$, so we have

$$\frac{d}{dt} (e^{\lambda t} \xi_t) = -\lambda e^{\lambda t} F_{N,\alpha}(x^\star - \xi_t, t).$$

Integrating from 0 to t gives

$$e^{\lambda t} \xi_t - \xi_0 = \int_0^t \frac{d}{ds} (e^{\lambda s} \xi_s) ds = -\lambda \int_0^t e^{\lambda s} F_{N,\alpha}(x^\star - \xi_s, s) ds.$$

Rewriting in terms of $\bar{x}_t^N$ yields the exact representation

$$\bar{x}_t^N = (1 - e^{-\lambda t}) x^\star + e^{-\lambda t} \bar{x}_0^N + \lambda e^{-\lambda t} \int_0^t e^{\lambda s} F_{N,\alpha}(\bar{x}_s^N, s) ds. \quad (3.6)$$

Here,

$$F_{N,\alpha}(z, s) = (z - x^\star) + e^{-\lambda s} \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(z + e^{-\lambda s} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda s} r_0^k))} \quad (3.7)$$

for some $z \in \mathbb{R}^D$.

We see that the particle average is composed of three contributions: 1. the term $(1 - e^{-\lambda t})x^\star$, converging exponentially in time toward the minimizer; 2. the term $e^{-\lambda t} \bar{x}_0^N$, decaying exponentially fast; 3. and finally the integral contribution, which encodes the dependence on both the concentration parameter α and the number of particles N .

Our aim is now to estimate how far the particle average $\bar{x}_t^N$ remains from the minimizer $x^\star$ for given parameters α and N . To this end, it is essential to understand the term $F_{N,\alpha}(\bar{x}_s^N, s)$, and in particular to derive upper bounds on its 2-norm.

This analysis is carried out in the next two subsections. In Section 3.1, we compare the term

$$\tilde{F}_{N,\alpha}(z, s) := \sum_{m=1}^N r_0^m \frac{\exp(-\alpha f(z + e^{-\lambda s} r_0^m))}{\sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda s} r_0^k))} \quad (3.8)$$

for $z \in [-K_\rho, K_\rho]^D$ and $s \geq 0$, with the limiting quantity

$$\tilde{F}_{\infty,\alpha}(z, s) := \frac{1}{\int_{[0,1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx} \int_{[0,1]^D} \mathbf{s}(x) \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx. \quad (3.9)$$

The restriction $z \in [-K, K]^D$ arises from the assumption that all initial points satisfy $x_0^n \in [-K, K]^D$, ensuring that the average remains in this set for every time t , since by [Koreaner], the convex hull of the particle system is contracting.

In Lemma 3.3, we will show

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s). \quad (3.10)$$

using classical tools from Quasi-Monte Carlo theory and assuming that all initial particle position arise from a sequence, which is uniformly distributed modulo one. We will additionally quantify the convergence rate in N . An important interpretation of this result is that $\tilde{F}_{N,\alpha}$ acts as a quadrature approximation of its infinite-particle limit $\tilde{F}_{\infty,\alpha}$.

In Section 3.2 we then analyse the limiting behaviour

$$\lim_{\alpha \rightarrow \infty} F_{\infty,\alpha}(z, s) \quad (3.11)$$

for $z \in [-K, K]^D$ and $s \geq 0$ and in Lemma 3.7 we will demonstrate that this term converges to zero, together with an explicit rate in α . The proof will rely on an application of Laplace's method.

3.1 Quasi-Monte Carlo Approximation for the Finite-Particle System

We first recall the notion of low-discrepancy point sets and collect several basic concepts and results that will be used throughout this section. Let $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$ be a set of N points. We define

$$A([\mathbf{a}, \mathbf{b}], \mathcal{P}) := \sum_{n=1}^N \mathbb{1}_{[\mathbf{a}, \mathbf{b}]}(\mathbf{p}^n),$$

where $\mathbb{1}_{[\mathbf{a}, \mathbf{b}]}$ denotes the indicator function of the interval $[\mathbf{a}, \mathbf{b}] = \prod_{i=1}^D [a_i, b_i]$. Hence, $A([\mathbf{a}, \mathbf{b}], \mathcal{P})$ counts the number of points of $\mathcal{P}$, which belong to $[\mathbf{a}, \mathbf{b}]$.

This counting function $A([\mathbf{a}, \mathbf{b}], \mathcal{P})$ provides the basis for the notion of *uniform distribution modulo one*, which formalizes the idea that the empirical distribution of a sequence of points converges to the uniform distribution on the unit cube.

Definition 3.1 (Uniform Distribution Modulo One). A sequence of points $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is called *uniformly distributed modulo one*, if

$$\lim_{N \rightarrow \infty} \left| \frac{A([\mathbf{a}, \mathbf{b}], \{\mathbf{p}^1, \dots, \mathbf{p}^N\})}{N} - \lambda_D([\mathbf{a}, \mathbf{b}]) \right| = 0$$

for all $\mathbf{a}, \mathbf{b} \in [0, 1]^D$ with $\mathbf{a} \leq \mathbf{b}$. Here, λ_D denotes the D -dimensional Lebesgue measure.

While Definition 3.1 describes the asymptotic behaviour of an infinite sequence, it is often convenient to quantify how well a finite point set approximates the uniform distribution. This leads to the related notion of discrepancy. For a point set $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$, the *discrepancy* is defined as

$$\mathcal{D}_N(\mathcal{P}) := \sup_{\substack{\mathbf{a}, \mathbf{b} \in [0, 1]^D \\ \mathbf{a} \leq \mathbf{b}}} \left| \frac{A([\mathbf{a}, \mathbf{b}], \mathcal{P})}{N} - \lambda_D([\mathbf{a}, \mathbf{b}]) \right|.$$

It measures the maximal deviation between the empirical distribution induced by $\mathcal{P}$ and the uniform distribution λ_D . A closely related variant is the *star-discrepancy*, which restricts the class of test sets to intervals starting at the origin. It is given by

$$\mathcal{D}_N^*(\mathcal{P}) := \sup_{\mathbf{a} \in [0, 1]^D} \left| \frac{A([0, \mathbf{a}], \mathcal{P})}{N} - \lambda_D([0, \mathbf{a}]) \right|. \quad (3.12)$$

The star-discrepancy plays a central role in Quasi-Monte Carlo integration. In particular, a sequence $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one if and only if

$$\lim_{n \rightarrow \infty} \mathcal{D}_N^*(\{\mathbf{p}^1, \dots, \mathbf{p}^N\}) = 0 \quad (3.13)$$

Point sets with small star-discrepancy are well suited for numerical integration. The following theorem summarizes two fundamental discrepancy-based error bounds for Quasi-Monte Carlo integration. The second inequality is the classical *Koksma-Hlawka* inequality, while the first provides a variant that applies to continuous integrands.

Theorem 3.2 (Koksma–Hlawka(-type) inequalities). Let $g : [0, 1]^D \rightarrow \mathbb{R}$ be continuous and let $\mathcal{P} = \{\mathbf{p}^1, \dots, \mathbf{p}^N\} \subset [0, 1]^D$ be a set of D -dimensional points. Then the following *Koksma–Hlawka-type* error bound holds:

$$\left| \frac{1}{N} \sum_{i=1}^N g(\mathbf{p}_i) - \int_{[0,1]^D} g(u) du \right| \leq 4w(g; \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)^{1/D}),$$

where

$$w(g; \delta) = \sup_{\substack{u, v \in [0,1]^D \\ \|u-v\|_\infty \leq \delta}} |g(u) - g(v)|, \quad \delta > 0$$

denotes the modulus of continuity of g . Moreover if g has bounded variation in the sense of Hardy–Krause, then the classical *Koksma–Hlawka* inequality

$$\left| \frac{1}{N} \sum_{i=1}^N g(\mathbf{p}_i) - \int_{[0,1]^D} g(u) du \right| \leq V(g) \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$$

holds.

Here, $V(g)$ denotes the *Hardy–Krause variation* of g . For sufficiently smooth functions $g \in C^{1, \dots, 1}(\mathbb{R}^D; \mathbb{R})$, this quantity is defined as follows: Let $k \in \{1, \dots, D\}$ and let $\mathcal{V} \subseteq \{1, \dots, D\}$ be an index set with $\#\mathcal{V} = k$. For $u \in [0, 1]^k$, we define the point $u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}} \in [0, 1]^D$ by

$$(u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}})_j = \begin{cases} u_j, & j \in \mathcal{V}, \\ 1, & j \notin \mathcal{V}. \end{cases}$$

Using this notation, we associate to g the k -dimensional function

$$g_{\mathcal{V}}(u) = g(u_{\mathcal{V}} : \mathbf{1}_{-\mathcal{V}}), \quad u \in [0, 1]^k.$$

For a function $f : [0, 1]^k \rightarrow \mathbb{R}$ that is k times continuously differentiable, we define its variation by

$$V(f) = \int_{[0,1]^k} \left| \frac{\partial^k}{\partial s_1 \cdots \partial s_k} f(s_1, \dots, s_k) \right| d(s_1, \dots, s_k),$$

given that the right-hand side exists. With these definitions, the Hardy–Krause variation of g is given by

$$V(g) = \sum_{k=1}^D \sum_{\substack{\mathcal{V} \subseteq \{1, \dots, D\} \\ \#\mathcal{V} = k}} V(g_{\mathcal{V}}).$$

Using these tools, we can now establish the following lemma, which relates $\tilde{F}_{N,\alpha}(z, t)$ and $\tilde{F}_{\infty,\alpha}(z, t)$.

Lemma 3.3. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be locally Lipschitz continuous and differentiable. Fix $z \in [-K, K]^D$, $\alpha \geq 0$, $\lambda > 0$ and $s \geq 0$. Assume that the sequence of points $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one. Moreover, let $\tilde{F}_{N,\alpha}(z, t)$ and $\tilde{F}_{\infty,\alpha}(z, t)$ be given by (3.8) and (3.9), respectively. Then

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s).$$

More precisely, the component-wise approximation error admits the quantitative bound

$$\left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| \leq 2\sqrt{D}K \left(\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N) + 4\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)^{1/D} \right) \frac{1 + C_{f,K} \alpha e^{-\lambda s}}{I(g_0) \exp(\alpha f(x^*))}$$

for any $d \in \{1, \dots, D\}$ with

$$I(g_0) = \int_{[0,1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx, \quad C_{f,K} = 2K \|\nabla f\|_\infty (1 + 2K).$$

Proof. First, for $d = 0, \dots, D$, we define the functions

$$g_d : [-2K, 2K]^D \rightarrow \mathbb{R}, \quad g_d(v) = v_d \exp(-\alpha f(z + e^{-\lambda s} v))$$

where $v = (v_1, \dots, v_D)^\top \in \mathbb{R}^D$ and we set $v_0 = 1$. Since f is locally Lipschitz continuous and bounded from below, the function g_d is locally Lipschitz on $\mathbb{R}^D$. When restricted to the compact domain $[-2K, 2K]^D$, it follows that g_d is bounded and globally Lipschitz with some constant $L_{g_d} > 0$.

Moreover, since f is differentiable, the function g_d is differentiable as well. An application of the mean value theorem therefore yields the existence of a point $\zeta \in [-2K, 2K]^D$ such that

$$|g_d(u) - g_d(v)| = \|\nabla g_d(\zeta)\| \|u - v\| \leq \sup_{w \in [-2K, 2K]^D} \|\nabla g_d(w)\| \|u - v\|.$$

By the chain rule, the gradient of g_d is given by

$$\nabla g_d(v) = e_d \exp(-\alpha f(z + v e^{-\lambda s})) - \alpha v_d e^{-\lambda s} \nabla f(z + v e^{-\lambda s}) \exp(-\alpha f(z + v e^{-\lambda s})),$$

where $e_0 = \mathbf{0}_D$ denotes the zero vector and e_d is the d -th canonical basis vector of $\mathbb{R}^D$.

This representation allows us to bound the Lipschitz constant associated with g_d as

$$\begin{aligned} L_{g_d} &\leq \sup_{w \in [-2K, 2K]^D} \|\exp(-\alpha f(z + w e^{-\lambda s}))(e_d - \alpha w_d e^{-\lambda s} \nabla f(z + w e^{-\lambda s}))\| \\ &\leq \sup_{w \in [-2K, 2K]^D} \exp(-\alpha f(z + w e^{-\lambda s})) \left(\|e_d\| + \alpha |w_d| e^{-\lambda s} \sup_{x \in [-2K, 2K]^D} \|\nabla f(x)\| \right). \end{aligned}$$

Since $w \in [-2K, 2K]^D$ and x^* is a global minimizer of f , we obtain for $d = 1, \dots, D$ the upper bound

$$L_{g_d} \leq \exp(-\alpha f(x^*)) (1 + 2\alpha K e^{-\lambda s} \|\nabla f\|_\infty) \quad (3.14)$$

and for $d = 0$ the estimate

$$L_{g_0} \leq 2 \exp(-\alpha f(x^*)) \alpha K e^{-\lambda s} \|\nabla f\|_\infty, \quad (3.15)$$

Here, the quantity

$$\|\nabla f\|_\infty = \sup_{x \in [-2K, 2K]^D} \|\nabla f(x)\|$$

denotes the supremum norm of the gradient of f over the relevant domain.

Next, we collect several observations concerning the mapping $\mathfrak{s}$, which will allow us to rewrite $\tilde{F}_{N,\alpha}(z, t)$ in a form that is well suited for a Quasi-Monte Carlo analysis. We begin by noting that the linearity of $\mathfrak{s}$ implies that the components

$$(\mathfrak{s}_d)_\mathcal{V}(u) = (2K u_d - K) \mathbb{1}_{\{d \in \mathcal{V}\}} + K \mathbb{1}_{\{d \notin \mathcal{V}\}}$$

for $d = 1, \dots, D$ admit particularly simple partial derivatives. Indeed, for $d' = 1, \dots, D$ we have

$$\frac{\partial}{\partial u_{d'}} (\mathfrak{s}_d)_{\{d'\}}(u) = 2K \mathbb{1}_{\{d=d'\}}, \quad \frac{\partial^k}{\partial u_1 \dots \partial u_k} (\mathfrak{s}_d)_\mathcal{V}(u) = 0$$

for all $1 < k \leq D$ and all subsets $\mathcal{V} \subseteq \{1, \dots, D\}$ with $\#\mathcal{V} > 1$. In particular, all mixed and higher-order derivatives vanish. As a consequence, the Hardy-Krause variation of $\mathfrak{s}_d$ reduces to

$$V(\mathfrak{s}_d) = \sum_{\substack{\mathcal{V} \subseteq \{1, \dots, D\} \\ \#\mathcal{V}=1}} V((\mathfrak{s}_d)_\mathcal{V}) = \sum_{d'=1}^D \int_0^1 \left| \frac{\partial}{\partial u_{d'}} (\mathfrak{s}_d)_{\{d'\}}(u) \right| du_{d'} = \int_0^1 2K du_d = 2K.$$

Furthermore, the linearity of $\mathfrak{s}$ also implies that the initial residual satisfies

$$r_0^{m,d} = x_0^{m,d} - \bar{x}_0^{N,d} = \mathfrak{s}_d(\mathbf{p}^m) - \frac{1}{N} \sum_{k=1}^N \mathfrak{s}_d(\mathbf{p}^k) = \mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)$$

for all $m = 1, \dots, N$ and $d = 1, \dots, D$. Substituting this identity into the definition of $\tilde{F}_{N,\alpha}(z, t)$ yields

$$\tilde{F}_{N,\alpha}(z, t) = \frac{1}{N} \sum_{m=1}^N (\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)) \frac{\exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}{\frac{1}{N} \sum_{k=1}^N \exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^k) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}.$$

By construction of the functions g_d , this expression can be written componentwise as

$$(\tilde{F}_{N,\alpha}(z, t))_d = \frac{\frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N))}{\frac{1}{N} \sum_{k=1}^N g_0(\mathfrak{s}(\mathbf{p}^k) - \mathfrak{s}(\bar{\mathbf{p}}^N))}, \quad (\tilde{F}_{\infty,\alpha}(z, s))_d := \frac{\int_{[0,1]^D} g_d(\mathfrak{s}(x)) dx}{\int_{[0,1]^D} g_0(\mathfrak{s}(x)) dx}.$$

for $d = 1, \dots, D$

We now introduce, for $d = 0, \dots, D$,

$$I(g_d) = \int_{[0,1]^D} g_d(\mathfrak{s}(x)) dx, \quad I'_N(g_d) = \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m)).$$

The absolute difference $|I(g_d) - I'_N(g_d)|$ can be estimated by the Koksma-Hlawka-type inequality. More precisely, applying Theorem 3.2 to the integrand $g_d \circ \mathfrak{s} : [0, 1]^D \rightarrow \mathbb{R}$ yields

$$|I(g_d) - I'_N(g_d)| \leq 4\omega(g_d \circ \mathfrak{s}; \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)^{1/D}),$$

where $\mathcal{D}_N^* = \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$ denotes the star discrepancy of the point set $\{\mathbf{p}_1, \dots, \mathbf{p}_N\}$. Note that the composition $g_d \circ \mathfrak{s}$ is continuous, since g_d is continuous and $\mathfrak{s}$ is linear. Moreover, the global Lipschitz continuity of g_d implies that the modulus of continuity of $g_d \circ \mathfrak{s}$ satisfies

$$\begin{aligned} \omega(g_d \circ \mathfrak{s}; (\mathcal{D}_N^*)^{1/D}) &= \sup_{\substack{u, v \in [0,1]^D \\ \|u-v\|_\infty \leq (\mathcal{D}_N^*)^{1/D}}} |g_d(\mathfrak{s}(u)) - g_d(\mathfrak{s}(v))| \\ &\leq \sup_{\substack{u, v \in [0,1]^D \\ \|u-v\|_\infty \leq (\mathcal{D}_N^*)^{1/D}}} L_{g_d} \|\mathfrak{s}(u) - \mathfrak{s}(v)\|. \end{aligned}$$

Next, exploiting again the linearity of $\mathfrak{s}$ and the fact that $\mathbf{p}^m$, and hence also $\bar{\mathbf{p}}^N$, take values in the unit cube $[0, 1]^D$, we observe that

$$|\mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)| = 2K |\mathbf{p}^{m,d} - \bar{\mathbf{p}}^{N,d}| \leq 2K. \quad (3.16)$$

Consequently, for arbitrary $u, v \in [0, 1]^D$, we obtain the norm bound

$$\|\mathfrak{s}(u) - \mathfrak{s}(v)\| \leq 2\sqrt{D}K.$$

Combining these estimates, the modulus of continuity of $g_d \circ \mathfrak{s}$ can be further bounded by

$$\omega(g_d \circ \mathfrak{s}; (\mathcal{D}_N^*)^{1/D}) \leq \sup_{\substack{u, v \in [0,1]^D \\ \|u-v\|_\infty \leq (\mathcal{D}_N^*)^{1/D}}} 2\sqrt{D}KL_{g_d}\|u-v\| \leq 2\sqrt{D}KL_{g_d}(\mathcal{D}_N^*)^{1/D},$$

where we used the equivalence of the norms on $\mathbb{R}^D$. Altogether, this yields the estimate

$$|I'_N(g_d) - I_N(g_d)| \leq 8\sqrt{D}KL_{g_d}(\mathcal{D}_N^*)^{1/D}. \quad (3.17)$$

We further define

$$I_N(g_d) = \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)).$$

Using the bound (3.16), we immediately obtain an estimate for the absolute ratio of $I_N(g_d)$ and $I_N(g_0)$, namely

$$\left| \frac{I_N(g_d)}{I_N(g_0)} \right| \leq \frac{\frac{1}{N} \sum_{m=1}^N |\mathfrak{s}_d(\mathbf{p}^m) - \mathfrak{s}_d(\bar{\mathbf{p}}^N)| \exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))}{\frac{1}{N} \sum_{m=1}^N \exp(-\alpha f(z + e^{-\lambda t}[\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)]))} \leq 2K. \quad (3.18)$$

Next, we estimate the absolute difference $|I'_N(g_d) - I_N(g_d)|$ by exploiting the Lipschitz continuity of L_{g_d} . Indeed, we compute

$$\begin{aligned} |I'_N(g_d) - I_N(g_d)| &= \left| \frac{1}{N} \sum_{m=1}^N g_d(\mathfrak{s}(\mathbf{p}^m)) - g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N)) \right| \\ &\leq \frac{1}{N} \sum_{m=1}^N |g_d(\mathfrak{s}(\mathbf{p}^m)) - g_d(\mathfrak{s}(\mathbf{p}^m) - \mathfrak{s}(\bar{\mathbf{p}}^N))| \\ &\leq \frac{1}{N} \sum_{m=1}^N L_{g_d} \|\mathfrak{s}(\bar{\mathbf{p}}^N)\| \end{aligned}$$

To bound the remaining term, we first observe that

$$\int_{[0,1]^D} \mathfrak{s}_d(x) dx = \int_{[0,1]^{D-1}} \int_0^1 2Kx_d - K dx_d dx_{-d} = \int_{[0,1]^{D-1}} 0 dx_{-d} = 0,$$

where x_{-d} denotes all components of the vector x except for the d -th one. Since $\mathfrak{s}_d$ is linear and hence continuous, and since its variation in the sense of Hardy-Krause satisfies $V(\mathfrak{s}_d) \leq 2K$, we may apply the Koksma-Hlawka inequality from Theorem 3.2 to obtain

$$\|\mathfrak{s}(\bar{\mathbf{p}}^N)\| = \sqrt{\sum_{d=1}^D \left| \frac{1}{N} \sum_{m=1}^N \mathfrak{s}_d(\mathbf{p}^m) - \int_{[0,1]^D} \mathfrak{s}_d(x) dx \right|^2} \leq \sqrt{\sum_{d=1}^D |2K\mathcal{D}_N^*|^2} = 2\sqrt{D}K\mathcal{D}_N^*.$$

Combining the above estimates yields

$$|I(g_d) - I'_N(g_d)| \leq 2\sqrt{D}KL_{g_d}\mathcal{D}_N^*. \quad (3.19)$$

With the construction of the quantities $I_N(g_d)$ and $I(g_d)$, we can bound the desired component-wise absolute difference between $\tilde{F}_{N,\alpha}(z, s)$ and $\tilde{F}_{\infty,\alpha}(z, s)$. More precisely, for a fixed component index $d \in \{1, \dots, D\}$, we obtain

$$\begin{aligned} |(\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d| &= \left| \frac{I_N(g_d)}{I_N(g_0)} - \frac{I(g_d)}{I(g_0)} \right| \\ &\leq \left| \frac{I_N(g_d)}{I(g_0)} - \frac{I(g_d)}{I(g_0)} \right| + \left| I_N(g_d) \left(\frac{1}{I_N(g_0)} - \frac{1}{I(g_0)} \right) \right| \\ &= \frac{|I_N(g_d) - I(g_d)|}{I(g_0)} + \left| \frac{I_N(g_d)}{I_N(g_0)} \right| \frac{|I(g_0) - I_N(g_0)|}{I(g_0)} \\ &\leq \frac{1}{I(g_0)} |I(g_d) - I_N(g_d)| + \frac{2K}{I(g_0)} |I(g_0) - I_N(g_0)| \end{aligned}$$

where in the last step we used the bound (3.18) on the ratio $|I_N(g_d)/I_N(g_0)|$.

Next, for $d = 0, \dots, D$, we combine the estimates obtained in (3.19) and (3.17) to control the difference between $I(g_d)$ and $I_N(g_d)$. Indeed, we have

$$|I(g_d) - I_N(g_d)| \leq |I(g_d) - I'_N(g_d)| + |I'_N(g_d) - I_N(g_d)| \leq 2\sqrt{D}KL_{g_d} \left(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D} \right).$$

Substituting this estimate into the previous bound yields

$$|(\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d| \leq \frac{2\sqrt{D}K(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D})}{I(g_0)} (L_{g_d} + 2KL_{g_0}).$$

We now insert the bounds for the Lipschitz constants L_{g_d} and L_{g_0} given in (3.14) and (3.15), respectively to obtain

$$\begin{aligned} (L_{g_d} + 2KL_{g_0}) &\leq \exp(-\alpha f(x^*)) (1 + 2\alpha K e^{-\lambda s} \|\nabla f\|_\infty + 4\alpha K^2 e^{-\lambda s} \|\nabla f\|_\infty) \\ &= \exp(-\alpha f(x^*)) (1 + 2\alpha K e^{-\lambda s} \|\nabla f\|_\infty (1 + 2K)) \\ &= \exp(-\alpha f(x^*)) (1 + C_{f,K} \alpha e^{-\lambda s}) \end{aligned}$$

where we introduce the constant $C_{f,K} = 2K\|\nabla f\|_\infty(1 + 2K)$.

Altogether, we therefore arrive at the estimate

$$\left|(\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d\right| \leq 2\sqrt{D}K \left(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D}\right) \frac{1 + C_{f,K}\alpha e^{-\lambda s}}{I(g_0) \exp(\alpha f(x^*))}.$$

Recall that our assumption on the uniform distribution modulo one of the points $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}}$ is equivalent to

$$\lim_{N \rightarrow \infty} \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N) = 0.$$

Consequently, for each component $d \in \{1, \dots, D\}$,

$$\lim_{N \rightarrow \infty} \left|(\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d\right| = 0,$$

which implies the convergence of the full vector-valued map

$$\lim_{N \rightarrow \infty} \tilde{F}_{N,\alpha}(z, s) = \tilde{F}_{\infty,\alpha}(z, s).$$

□

The estimate in Lemma 3.3 shows that the dependence on the number of points N acts only through the star discrepancy $\mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$. As the sequence of points $\mathcal{S}$ is uniformly distributed modulo one, this quantity converges to zero as $N \rightarrow \infty$. Hence, $\tilde{F}_{N,\alpha}(z, s)$ converges to $\tilde{F}_{\infty,\alpha}(z, s)$ as $N \rightarrow \infty$. The convergence rate becomes slower with increasing dimension D , reflecting the well-known curse of dimensionality in Quasi-Monte Carlo methods.

The concentration parameter α enters the bound both through the denominator $I(g_0) \exp(\alpha f(\hat{x}))$ and linearly through the factor $1 + C_{f,K}\alpha e^{-\lambda s}$. Its influence will be analyzed in the next section. All remaining constants depend only on the dimension D , K and the regularity of f through $\|\nabla f\|_\infty$.

3.2 Laplace Method for the Finite-Concentration System

We now analyze the finite-concentration system by investigating the asymptotic behavior of $F_{\infty,\alpha}(z, s)$ as $\alpha \rightarrow \infty$. Our approach is based on Laplace methods, which allow us to rigorously quantify how integrals weighted by exponentially concentrated densities localize around the global minimizer of the objective function.

We first introduce some notation. The open D -dimensional cube with center $x \in \mathbb{R}^D$ and half side-length $r > 0$ is denoted by

$$B_r^\infty(x) := \{y \in \mathbb{R}^D : \|y - x\|_\infty < r\} = (x - r\mathbf{1}_D, x + r\mathbf{1}_D) = \prod_{d=1}^D (x_d - r, x_d + r). \quad (3.20)$$

and we denote its closure by

$$\overline{B}_r^\infty(x) := \prod_{d=1}^D [x_d - r, x_d + r]. \quad (3.21)$$

A key ingredient in controlling each component of $F_{\infty,\alpha}(z, s)$ is the integrability of polynomials and exponentially decaying weights. By a change of variables in polar coordinates, one obtains the identity

$$\int_{\mathbb{R}^D} \|x\|^b \exp(-a\|x\|^\beta) dx = \frac{2}{\beta} \pi^{\frac{D}{2}} a^{-\frac{D+b}{\beta}} \frac{\Gamma((D+b)/\beta)}{\Gamma(D/2)}, \quad (3.22)$$

which is finite for any $a > 0$ and $b \geq 0$.

To demonstrate the convergence of $F_{\infty,\alpha}(z, s)$ to $\mathbf{0}_D$ as $\alpha \rightarrow \infty$, we impose the following assumptions on the objective function f and its minimizer x^* .

Assumption 3.4. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be given and let $x^* \in \mathbb{R}^D$. We assume that the following conditions hold.

- (A1) **Global minimality and uniqueness.** The point x^* is a unique global minimizer of f . Moreover, for every $\delta > 0$ there exists a constant $c = c(\delta) > 0$ such that

$$f(x) \geq f(x^*) + c$$

for every $x \notin B_\delta^\infty(x^*)$.

(A2) **Local growth condition near the minimizer.** There exist constants $\delta_{\text{loc}} > 0$, $m_f > 0$, $C_f > 0$, and exponents $\beta > 0$, $\gamma > 0$ such that

$$f(x) = f(x^*) + m_f \|x - x^*\|^\beta + R(x, x^*)$$

for all $x \in B_{\delta_{\text{loc}}}^\infty(x^*)$, where the remainder term R satisfies

$$|R(x, x^*)| \leq C_f \|x - x^*\|^{\beta+\gamma}.$$

By possibly reducing δ_{loc} , we may further assume that

$$|R(x, x^*)| \leq \frac{m_f}{2} \|x - x^*\|^\beta$$

for all $x \in B_{\delta_{\text{loc}}}^\infty(x^*)$.

(A3) **Boundedness of the minimizer.** There exists a constant $\rho \in (0, 1)$ such that

$$x^* \in B_{K\rho}^\infty(\mathbf{0}_D), \quad K_\rho := K\rho.$$

Assumption (A1) ensures that $B_{\delta_{\text{loc}}}^\infty(x^*)$ is the unique region in which the global minimum value of f is attained and that f is separated from this minimum outside any neighborhood of x^* . Assumption (A2) provides a quantitative control of the local behavior of f around the minimizer. The growth exponents β and γ determine the sharpness of the minimum. Finally, Assumption (A3) ensures that the minimizer lies in an open cube in the state space from which the initial particle positions $x_0^1, \dots, x_0^N$ are drawn.

Before analysing the limit $\alpha \rightarrow \infty$, we first recall the relation between the finite-particle system and its limit $N \rightarrow \infty$. Inserting (3.8) into (3.7) yields

$$F_{N,\alpha}(z, s) = (z - x^*) + e^{-\lambda s} \tilde{F}_{N,\alpha}(z, s), \quad (3.23)$$

for all $z \in [-K, K]^D$ and $s \geq 0$. By Lemma 3.3, the limit as $N \rightarrow \infty$ is given by

$$F_{\infty,\alpha}(z, s) := \lim_{N \rightarrow \infty} F_{N,\alpha}(z, s) = (z - x^*) + e^{-\lambda s} \tilde{F}_{\infty,\alpha}(z, s), \quad (3.24)$$

where $\tilde{F}_{\infty,\alpha}(z, s)$ is defined in (3.9).

In order to establish convergence of the finite system, it suffices to analyze the asymptotic behavior of $F_{\infty,\alpha}(z, s)$ as $\alpha \rightarrow \infty$.

Lemma 3.5. Suppose that Assumptions 3.4 hold, and let $F_{\infty,\alpha}(z, s)$ be defined by (3.24). Assume that there exists a radius $\delta \leq \delta_{\text{loc}}$ such that

$$B_\delta^\infty(x^*) \subseteq B_{Ke^{-\lambda s}}^\infty(z)$$

for some $z \in [-K, K]^D$ and $s \geq 0$. Then

$$\lim_{\alpha \rightarrow \infty} F_{\infty,\alpha}(z, s) = \mathbf{0}_D.$$

More precisely, the component-wise approximation error admits, for $d \in \{1, \dots, D\}$, the quantitative bound

$$|(F_{\infty,\alpha}(z, s))_d| \leq \frac{\alpha^{-\frac{1+\gamma}{\beta}}}{C_0(\alpha)} C_{f,D,\beta,\gamma} + \frac{\alpha^{\frac{D}{\beta}}}{C_0(\alpha) \exp(\alpha c)} (2K)^{D+1} e^{-\lambda s(D+1)},$$

where $C_{f,D,\beta,\gamma}$ is a constant depending only on the objective f , the dimension D , β and γ . Moreover, the normalization factor $C_0(\alpha)$ converges, as $\alpha \rightarrow \infty$, to a positive constant depending only on f , D and β , but not on α .

Proof. By (3.24), we already know that the d -th component of $F_{\infty,\alpha}(z, s)$ can be written as

$$\begin{aligned} (F_{\infty,\alpha}(z, s))_d &= (z_d - x_d^*) + e^{-\lambda s} \frac{\int_{[0,1]^D} \mathbf{s}_d(x) \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx}{\int_{[0,1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx} \\ &= \frac{\int_{[0,1]^D} (z_d + e^{-\lambda s} \mathbf{s}_d(x) - x_d^*) \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx}{\int_{[0,1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathbf{s}(x))) dx}. \end{aligned}$$

To simplify the structure of the integrals, we consider the affine change of variables

$$y = z + e^{-\lambda s} \mathfrak{s}(x).$$

Since $\mathfrak{s}$ is a linear mapping from $[0, 1]^D$ onto $[-K, K]^D$, it follows that

$$\mathbf{0}_D \leq x \leq \mathbf{1}_D \Leftrightarrow -Ke^{-\lambda s} \mathbf{1}_D \leq e^{-\lambda s} \mathfrak{s}(x) \leq Ke^{-\lambda s} \mathbf{1}_D \Leftrightarrow y \in \overline{B}_{Ke^{-\lambda s}}^\infty(z).$$

In particular, the mapping

$$\Phi : [0, 1]^D \rightarrow \overline{B}_{Ke^{-\lambda s}}^\infty(z), \quad \Phi(x) := z + e^{-\lambda s} \mathfrak{s}(x).$$

is continuously differentiable on $[0, 1]^D$. Moreover, the Jacobian of Φ is constant and given by

$$D\Phi(x) = e^{-\lambda s} D\mathfrak{s} = 2Ke^{-\lambda s} I_D,$$

where I_D denotes the D -dimensional identity matrix. Consequently, the determinant of the Jacobian is constant and equals

$$\det(D\Phi(x)) = (2K)^D e^{-D\lambda s}.$$

Since $\mathfrak{s}$ is bijective between $[0, 1]^D$ and $[-K, K]^D$, the mapping Φ is a bijective as well, and its inverse is continuous differentiable. Hence, Φ is a diffeomorphism in the sense required for a change of variables. Applying said change of variables yields

$$\int_{[0, 1]^D} \frac{z_d + e^{-\lambda s} \mathfrak{s}_d(x) - x_d^*}{\exp(\alpha f(z + e^{-\lambda s} \mathfrak{s}(x)))} dx = \frac{1}{(2K)^D e^{-D\lambda s}} \int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} (y_d - x_d^*) \exp(-\alpha f(y)) dy$$

and similarly

$$\int_{[0, 1]^D} \exp(-\alpha f(z + e^{-\lambda s} \mathfrak{s}(x))) dx = \frac{1}{(2K)^D e^{-D\lambda s}} \int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy.$$

As a result, the expression for $(F_{\infty, \alpha}(z, s))_d$ simplifies to

$$(F_{\infty, \alpha}(z, s))_d = \frac{\int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} (y_d - x_d^*) \exp(-\alpha f(y)) dy}{\int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy}. \quad (3.25)$$

By assumption, there exists a $\delta \leq \delta_{\text{loc}}$ such that $B_\delta^\infty(x^*) \subseteq \overline{B}_{Ke^{-\lambda s}}^\infty(z)$. Fixing such a δ , we define

$$D_R(z, s) := \overline{B}_{Ke^{-\lambda s}}^\infty(z) \setminus B_\delta^\infty(x^*).$$

We now split the domain of integration in (3.25) into the disjoint subsets $B_\delta^\infty(x^*)$ and $D_R(z, s)$, and investigate the resulting integrals separately. This decomposition separates the contribution coming from a neighborhood of the minimizer x^* from the contribution away from the minimizer.

We first focus on the denominator of (3.25), which is given by

$$\int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy = \int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy + \int_{D_R(z, s)} \exp(-\alpha f(y)) dy.$$

Since the exponential function is strictly positive, the second term on the right-hand side is strictly larger than zero. Consequently, we obtain the lower bound

$$\int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} \exp(-\alpha f(y)) dy \geq \int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy.$$

We now analyze the remaining integral over $B_\delta^\infty(x^*)$. Since $\delta \leq \delta_{\text{loc}}$, assumption (A2) applies, and we can write

$$\int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy = \exp(-\alpha f(x^*)) \int_{B_\delta^\infty(x^*)} \exp(-\alpha m_f \|y - x^*\|^\beta - \alpha R(y, x^*)) dy.$$

This representation isolates the exponential decay around the minimizer $x^\star$. To make the asymptotic behavior as $\alpha \rightarrow \infty$ explicit, we perform a change of variables that rescales the neighborhood of $x^\star$. We introduce the affine transformation

$$w = \alpha^{\frac{1}{\beta}}(y - x^\star) \quad \Leftrightarrow \quad y = x^\star + \alpha^{-\frac{1}{\beta}}w,$$

under which the integration domain transforms as

$$y \in B_\delta^\infty(x^\star) \quad \Leftrightarrow \quad x^\star + \alpha^{-\frac{1}{\beta}}w \in B_\delta^\infty(x^\star) \quad \Leftrightarrow \quad w \in B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D).$$

Motivated by this rescaling, we define the mapping

$$\Psi : B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D) \rightarrow B_\delta^\infty(x^\star), \quad \Psi(w) := x^\star + \alpha^{-\frac{1}{\beta}}w.$$

Both $B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)$ and $B_\delta^\infty(x^\star)$ are open sets. Since Ψ is affine, it is a diffeomorphism with continuous differentiable inverse and constant Jacobian

$$D\Psi(w) = \alpha^{-\frac{1}{\beta}}I_D,$$

whose determinant is given by

$$|\det D\Psi(w)| = \alpha^{-\frac{D}{\beta}}.$$

Using the identity

$$\|y - x^\star\|^\beta = \left\|x^\star + \alpha^{-\frac{1}{\beta}}w - x^\star\right\|^\beta = \alpha^{-1}\|w\|^\beta$$

a change of variables yields

$$\int_{B_\delta^\infty(x^\star)} \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D}{\beta}}}{\exp(\alpha f(x^\star))} \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) dw.$$

For convenience, we introduce the quantity

$$C_0(\alpha) := \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) dw. \quad (3.26)$$

With this notation, we obtain

$$\int_{B_\delta^\infty(x^\star)} \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D}{\beta}}}{\exp(\alpha f(x^\star))} C_0(\alpha).$$

Before turning to the numerator of (3.25), we further analyze the behavior of $C_0(\alpha)$. By monotonicity of the exponential function, Assumption (A2), and the fact that $\alpha > 0$, we obtain the upper bound

$$\begin{aligned} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) &\leq \exp\left(-m_f\|w\|^\beta + \alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right|\right) \\ &\leq \exp\left(-m_f\|w\|^\beta + \alpha \frac{m_f}{2} \left\|\alpha^{-\frac{1}{\beta}}w\right\|^\beta\right) \\ &= \exp\left(-\frac{m_f}{2}\|w\|^\beta\right). \end{aligned}$$

Analogously, we obtain the lower bound

$$\begin{aligned} \exp\left(-m_f\|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right) &\geq \exp\left(-m_f\|w\|^\beta - \alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}}w, x^\star\right)\right|\right) \\ &= \exp\left(-\frac{3m_f}{2}\|w\|^\beta\right). \end{aligned}$$

Together, these inequalities imply the bounds.

$$\int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-\frac{3m_f}{2}\|x\|^\beta\right) dx \leq C_0(\alpha) \leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} \exp\left(-\frac{m_f}{2}\|x\|^\beta\right) dx. \quad (3.27)$$

We now study the asymptotic behavior of $C_0(\alpha)$ as $\alpha \rightarrow \infty$ using the dominated convergence theorem. To this end, we rewrite

$$C_0(\alpha) = \int_{\mathbb{R}^D} \underbrace{\exp\left(-m_f \|w\|^\beta - \alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right) \mathbb{1}_{B_{\delta\alpha^{1/\beta}}(\mathbf{0}_D)}(w)}_{=: g_\alpha(w)} dw.$$

By the bounds in (3.27) and the positivity of the exponential function, the integrand g_α is dominated by

$$\exp\left(-\frac{m_f}{2} \|w\|^\beta\right) \mathbb{1}_{\mathbb{R}^D}(w),$$

which is integrable. Indeed, using (3.22) with $b = 0$ and $a = \frac{m_f}{2}$, we obtain

$$\int_{\mathbb{R}^D} \exp\left(-\frac{m_f}{2} \|w\|^\beta\right) dw = \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D}{\beta}} \frac{\Gamma(D/\beta)}{\Gamma(D/2)} < \infty.$$

Moreover, Assumption (A2) implies that

$$\begin{aligned} \exp\left(-\alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right) &\leq \exp\left(\alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right|\right) \\ &\leq \exp\left(\alpha C_f \left\|\alpha^{-\frac{1}{\beta}} w\right\|^{\beta+\gamma}\right) \\ &= \exp\left(C_f \alpha^{-\frac{\gamma}{\beta}} \|w\|^{\beta+\gamma}\right), \end{aligned}$$

and similarly,

$$\begin{aligned} \exp\left(-\alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right) &\geq \exp\left(-\alpha \left|R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right|\right) \\ &\geq \exp\left(-C_f \alpha^{-\frac{\gamma}{\beta}} \|w\|^{\beta+\gamma}\right). \end{aligned}$$

Since $\beta, \gamma > 0$, both bounds converge point-wise to $\exp(0) = 1$ as $\alpha \rightarrow \infty$. Therefore we have

$$\exp\left(-\alpha R\left(x^\star + \alpha^{-\frac{1}{\beta}} w; x^\star\right)\right) \rightarrow 1.$$

Together with the fact that

$$\mathbb{1}_{B_{\delta\alpha^{1/\beta}}(\mathbf{0}_D)}(w) \rightarrow \mathbb{1}_{\mathbb{R}^D}(w)$$

this yields the point-wise convergence

$$g_\alpha(x) \rightarrow \exp(-m_f \|w\|^\beta) \mathbb{1}_{\mathbb{R}^D}(w).$$

We may therefore apply the dominated convergence theorem to conclude that

$$C_0(\infty) = \int_{\mathbb{R}^D} \lim_{\alpha \rightarrow \infty} g_\alpha(w) dw = \int_{\mathbb{R}^D} \exp(-m_f \|w\|^\beta) dw = \frac{2}{\beta} \pi^{\frac{D}{2}} m_f^{-\frac{D}{\beta}} \frac{\Gamma((D)/\beta)}{\Gamma(D/2)}, \quad (3.28)$$

where the final identity follows from (3.22) with $a = m_f$ and $b = 0$.

We now determine analogous results for the numerator of (3.25). As before, we split the domain of integration into the disjoint subsets $B_\delta^\infty(x^\star)$ and $D_R(z, s)$. We first consider the integral over the neighborhood of the minimizer,

$$\int_{B_\delta^\infty(x^\star)} (y_d - x_d^\star) \exp(-\alpha f(y)) dy.$$

Using Assumption (A2) and applying the change of variables induced by the mapping Ψ , we obtain

$$\int_{B_\delta^\infty(x^\star)} (y_d - x_d^\star) \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D+1}{\beta}}}{\exp(\alpha f(x^\star))} C_1(\alpha), \quad (3.29)$$

where

$$C_1(\alpha) := \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp\left(-m_f \|w\|^\beta - \alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w, x^*\right)\right) dw. \quad (3.30)$$

As in the analysis of $C_0(\alpha)$, we derive bounds on $C_1(\alpha)$ and determine its limit as $\alpha \rightarrow \infty$. We begin by estimating $|C_1(\alpha)|$. By the mean value theorem, for any $v \in \mathbb{R}$ there exists ζ between 0 and v , such that for the continuous function $h(u) = \exp(-u)$,

$$|\exp(-v) - 1| = |\exp(-v) - \exp(-0)| = \exp(-\zeta)|v|.$$

Since the exponential function is monotonically increasing and ζ lies between 0 and v , this yields

$$|\exp(-v) - 1| \leq \exp(|v|)|v|.$$

Next, we exploit the symmetry of the integration domain of $C_1(\alpha)$ and of the function $\exp(-m_f \|w\|^\beta)$. Each component w_d ranges over the symmetric interval $[-\delta\alpha^{1/\beta}, \delta\alpha^{1/\beta}]$. Writing $w = (w_d, w_{-d})$ and applying Fubini, we obtain

$$\int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp(-m_f \|w\|^\beta) dw = \int_{-\delta\alpha^{1/\beta}}^{\delta\alpha^{1/\beta}} \dots \int_{-\delta\alpha^{1/\beta}}^{\delta\alpha^{1/\beta}} w_d \exp(-m_f \|w\|^\beta) dw_d dw_{-d}.$$

For fixed w_{-d} , the factor $\exp(-m_f \|w\|^\beta)$ depends on w_d only through $|w_d|$. Hence, the contribution for w_d and $-w_d$ cancel, and the inner integral vanishes. Consequently,

$$\int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp(-m_f \|w\|^\beta) dw = 0.$$

Using this identity and the inequality above, we may write

$$\begin{aligned} |C_1(\alpha)| &= \left| \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} w_d \exp(-m_f \|w\|^\beta) \left(\exp\left(-\alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right)\right) - 1 \right) dw \right| \\ &\leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} |w_d| \exp(-m_f \|w\|^\beta) \left| \exp\left(-\alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right)\right) - 1 \right| dw \\ &\leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} |w_d| \exp(-m_f \|w\|^\beta) \exp\left(\alpha \left| R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right) \right| \right) \alpha \left| R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right) \right| dw \end{aligned}$$

Using assumption (A2), the fact that all terms are nonnegative, and estimating $|w_d| \leq \|w\|$, we further compute

$$\begin{aligned} |C_1(\alpha)| &\leq \int_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)} |w_d| \exp(-m_f \|w\|^\beta) \exp\left(\frac{m_f}{2} \|w\|^\beta\right) \alpha^{-\frac{\gamma}{\beta}} C_f \|w\|^{\beta+\gamma} dw \\ &\leq C_f \alpha^{-\frac{\gamma}{\beta}} \int_{\mathbb{R}^D} \|w\|^{1+\beta+\gamma} \exp\left(-\frac{m_f}{2} \|w\|^\beta\right) dw. \end{aligned}$$

Applying (3.22) with $a = \frac{m_f}{2}$ and $b = 1 + \beta + \gamma$, yields the final bound

$$|C_1(\alpha)| \leq C_f \alpha^{-\frac{\gamma}{\beta}} \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D+1+\beta+\gamma}{\beta}} \frac{\Gamma((D+1+\beta+\gamma)/\beta)}{\Gamma(D/2)}. \quad (3.31)$$

We now determine the limit of $C_1(\infty)$ as $\alpha \rightarrow \infty$. As before, we write

$$C_1(\alpha) = \int_{\mathbb{R}^D} \underbrace{w_d \exp\left(-m_f \|w\|^\beta - \alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w, x^*\right)\right)}_{=: \tilde{g}_\alpha(w)} \mathbb{1}_{B_{\delta\alpha^{1/\beta}}^\infty(\mathbf{0}_D)}(w) dw$$

By (3.27) and the estimate $|w_d| \leq \|w\|$, the integrand $\tilde{g}_\alpha$ is dominated by

$$\|w\| \exp\left(-\frac{m_f}{2} \|w\|^\beta\right) \mathbb{1}_{\mathbb{R}^D}(w),$$

which is integrable by (3.22) with $b = 1$ and $a = \frac{m_f}{2}$. Indeed, we have

$$\int_{\mathbb{R}^D} \|w\| \exp\left(-\frac{m_f}{2}\|w\|^\beta\right) dw = \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D+1}{\beta}} \frac{\Gamma((D+1)/\beta)}{\Gamma(D/2)} < \infty.$$

As shown previously, we have the point-wise convergence

$$\exp\left(-\alpha R\left(x^* + \alpha^{-\frac{1}{\beta}} w; x^*\right)\right) \rightarrow 1$$

as $\alpha \rightarrow \infty$, and therefore

$$\tilde{g}_\alpha(w) \rightarrow w_d \exp(-m_f \|w\|^\beta) \mathbf{1}_{\mathbb{R}^D}(w).$$

Applying the dominated convergence theorem yields

$$C_1(\infty) = \int_{\mathbb{R}^D} \lim_{\alpha \rightarrow \infty} \tilde{g}_\alpha(w) dw = \int_{\mathbb{R}^D} w_d \exp(-m_f \|w\|^\beta) dw = 0, \quad (3.32)$$

where the last identity follows from the symmetry of the integrand and the integration domain.

It remains to estimate the contribution of the numerator over the remainder set $D_R(z, s)$,

$$\int_{D_R(z, s)} (y_d - x_d^*) \exp(-\alpha f(y)) dy.$$

Since $|y_d - x_d^*|$ is non-negative and integrable on the compact, path-connected set $\overline{B}_{Ke^{-\lambda s}}^\infty(z) \setminus B_\delta^\infty(x^*)$ and $\exp(-\alpha f(y))$ is continuous, the mean value theorem for integrals implies that there exists a $\zeta \in D_R(z, s)$ such that

$$\begin{aligned} \left| \int_{D_R(z, s)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| &\leq \int_{D_R(z, s)} |y_d - x_d^*| \exp(-\alpha f(y)) dy \\ &= \exp(-\alpha f(\zeta)) \int_{D_R(z, s)} |y_d - x_d^*| dy. \end{aligned}$$

Since ζ lies outside the δ -neighborhood of the minimizer x^* , assumption (A1) yields

$$\left| \int_{D_R(z, t)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| \leq \exp(-\alpha f(x^*)) \exp(-\alpha c) \int_{\overline{B}_{Ke^{-\lambda s}}^\infty(z)} |y_d - x_d^*| dy,$$

where in the last step we enlarged the domain of integration, which is admissible since the integrand is non-negative.

Since $x^* \in B_{K_\rho e^{-\lambda s}}^\infty(z) \subseteq \overline{B}_{Ke^{-\lambda s}}^\infty(z)$, we have

$$|y_d - x_d^*| \leq 2Ke^{-\lambda s}$$

for all $y \in \overline{B}_{Ke^{-\lambda s}}^\infty(z)$, and using that the volume of the cube is $(2Ke^{-\lambda s})^D$, we conclude

$$\left| \int_{D_R(z, t)} (y_d - x_d^*) \exp(-\alpha f(y)) dy \right| \leq \exp(-\alpha f(x^*)) \exp(-\alpha c) (2K)^{D+1} e^{-\lambda s(D+1)}. \quad (3.33)$$

We now have all ingredients in place to conclude the proof by combining the previous estimates. Recalling the decomposition of numerator and denominator and using the lower bound on the denominator derived above, we obtain

$$\begin{aligned} (F_{\infty, \alpha}(z, s))_d &= \frac{\int_{B_\delta^\infty(x^*)} (y_d - x_d^*) \exp(-\alpha f(y)) dy + \int_{D_R(z, s)} (y_d - x_d^*) \exp(-\alpha f(y)) dy}{\int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy + \int_{D_R(z, s)} \exp(-\alpha f(y)) dy} \\ &\leq \frac{\int_{B_\delta^\infty(x^*)} (y_d - x_d^*) \exp(-\alpha f(y)) dy + \int_{D_R(z, s)} (y_d - x_d^*) \exp(-\alpha f(y)) dy}{C_0(\alpha) \exp(-\alpha f(x^*)) \alpha^{-\frac{D}{\beta}}}. \end{aligned}$$

Here, $C_0(\alpha)$ is defined in (3.26), satisfies the bounds in (3.27), and converges to a finite, positive constant according to (3.28). Applying the estimates (3.29) for the contribution from $B_\delta^\infty(x^*)$ and (3.33) for the contribution from $D_R(z, s)$, we obtain

$$\begin{aligned} |(F_{\infty, \alpha}(z, s))_d| &\leq \frac{\exp(-\alpha f(\hat{x}))\alpha^{-\frac{D+1}{\beta}}|C_1(\alpha)| + \exp(-\alpha f(x^*))\exp(-\alpha c)(2K)^{D+1}e^{-\lambda s(D+1)}}{C_0(\alpha)\exp(-\alpha f(x^*))\alpha^{-\frac{D}{\beta}}} \\ &\leq \frac{\alpha^{-\frac{1}{\beta}}|C_1(\alpha)|}{C_0(\alpha)} + \frac{\exp(-\alpha c)}{C_0(\alpha)\alpha^{-\frac{D}{\beta}}}(2K)^{D+1}e^{-\lambda s(D+1)}. \end{aligned}$$

Using the bound (3.31) on $|C_1(\alpha)|$, we finally arrive at

$$|(F_{\infty, \alpha}(z, s))_d| \leq \frac{\alpha^{-\frac{1+\gamma}{\beta}}}{C_0(\alpha)}\tilde{C}_{f, D, \beta, \gamma} + \frac{\alpha^{\frac{D}{\beta}}}{C_0(\alpha)\exp(\alpha c)}(2K)^{D+1}e^{-\lambda s(D+1)},$$

where

$$\tilde{C}_{f, D, \beta, \gamma} := C_f \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2}\right)^{-\frac{D+1+\beta+\gamma}{\beta}} \frac{\Gamma((D+1+\beta+\gamma)/\beta)}{\Gamma(D/2)}.$$

By (3.28), the quantity $C_0(\alpha)$ converges to a finite constant as $\alpha \rightarrow \infty$. Moreover, since the exponential term $\exp(\alpha c)$ grows faster than any polynomial in α , both terms on the right-hand side converge to zero. Combining all components, we conclude that

$$\lim_{\alpha \rightarrow \infty} F_{\infty, \alpha}(z, s) = \mathbf{0}_D.$$

□

The estimate in Lemma 3.7 shows that the asymptotic behaviour of $F_{\infty, \alpha}(z, s)$ is bounded by two different contributions. The first term decays polynomially in α at rate $\alpha^{-(D+\gamma)/\beta}$ and reflects the local approximation of the objective function f in a neighborhood of the minimizer x^* . The exponent explicitly depends on the dimension D , indicating that the convergence becomes faster as the dimension increases.

The second term is exponentially small in α and captures the contribution from outside the local neighbourhood of x^* . Its decay rate is dominated by the factor $\exp(-\alpha c)$, while the remaining prefactor depends only polynomially on α and on the dimension through K^{D+1} . Hence, for sufficiently large α , this term becomes negligible compared to the leading polynomial contribution.

The normalization factor $C_0(\alpha)$ converges to a positive constant as $\alpha \rightarrow \infty$ and therefore does not affect the asymptotic convergence rate. Overall, the lemma establishes that $F_{\infty, \alpha}(z, s)$ converges to zero as $\alpha \rightarrow \infty$, with a convergence rate that increases with increasing dimension D .

3.3 Combined Convergence Analysis of QMC²BO

In this section, we combine the quasi-Monte Carlo approximation from Lemma 3.3 with the Laplace method estimate from Lemma 3.7 in order to derive the convergence of the average particle position $\bar{x}_t^N$ of the QMC²BO dynamics towards the minimizer x^* of the objective function f . In particular, we obtain an explicit convergence rate that depends on both the number of particles N and the concentration parameter α .

To this end, we study the difference between the average particle position $\bar{x}_t^N$ and the minimizer x^* . According to (3.6), this difference admits the representation

$$\bar{x}_t^N - x^* = e^{-\lambda t} (\bar{x}_0^N - x^*) + \lambda e^{-\lambda t} \int_0^t e^{\lambda s} F_{N, \alpha}(\bar{x}_s^N, s) ds. \quad (3.34)$$

Here, $F_{N, \alpha}(\bar{x}_s^N, s)$ is defined as in (3.7). The first term on the right-hand side describes the exponentially decaying influence of the initial configuration, while the second term accounts for the cumulative effect of the interaction force. Consequently, controlling the convergence behavior of $\bar{x}_t^N - x^*$ reduces to obtaining suitable bounds on this integral term. Using (3.7) and (3.8), we can rewrite

$$\begin{aligned} |(F_{N, \alpha}(\bar{x}_s^N, s))_d| &= \left| (\bar{x}_s^N - x^*)_d + e^{-\lambda s} (\tilde{F}_{N, \alpha}(\bar{x}_s^N, s))_d \right| \\ &= \left| (\bar{x}_s^N - x^*)_d + e^{-\lambda s} (\tilde{F}_{\infty, \alpha}(\bar{x}_s^N, s))_d + e^{-\lambda s} (\tilde{F}_{N, \alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty, \alpha}(\bar{x}_s^N, s))_d \right| \\ &= \left| (F_{\infty, \alpha}(\bar{x}_s^N, s))_d + e^{-\lambda s} (\tilde{F}_{N, \alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty, \alpha}(\bar{x}_s^N, s))_d \right|, \end{aligned}$$

where, in the last step, we employ identity (3.24). This decomposition separates the quasi-Monte Carlo approximation error from the error that can be bounded by the Laplace method.

Under the assumptions of Lemma 3.3, the quasi-Monte Carlo error can be bounded by

$$\left| (\tilde{F}_{N,\alpha}(\bar{x}_s^N, s) - \tilde{F}_{\infty,\alpha}(\bar{x}_s^N, s))_d \right| \leq 2\sqrt{D}K \left(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D} \right) \frac{1 + 2K(1 + 2K)\|\nabla f\|_\infty \alpha e^{-\lambda s}}{I(g_0) \exp(\alpha f(\hat{x}))},$$

where $\mathcal{D}_N^* = \mathcal{D}_N^*(\mathbf{p}_1, \dots, \mathbf{p}_N)$ denotes the star-discrepancy of the point set $\{\mathbf{p}_1, \dots, \mathbf{p}_N\} \subset [0, 1]^D$. Moreover, from the proof of Lemma 3.7, we recall that, after an appropriate change of variables, the denominator appearing in the quasi-Monte Carlo error bound admits the lower bound

$$\begin{aligned} I(g_0) &= \int_{[0,1]^D} \exp(-\alpha f(\bar{x}_s^N + e^{-\lambda s} \mathbf{s}(x))) dx = \int_{\bar{B}_{Ke^{-\lambda s}(\bar{x}_s^N)}^\infty} \exp(-\alpha f(y)) dy \\ &\geq \int_{B_\delta^\infty(x^*)} \exp(-\alpha f(y)) dy = \frac{\alpha^{-\frac{D}{\beta}} C_0(\alpha)}{\exp(\alpha f(x^*))}. \end{aligned}$$

Combining the above estimates and using the elementary inequality $e^{-x} \leq 1$ for all $x \geq 0$, we arrive at the bound

$$\left| (\tilde{F}_{N,\alpha}(z, s) - \tilde{F}_{\infty,\alpha}(z, s))_d \right| \leq \mathbf{C}_{\text{QMC}}(N, \alpha)$$

for each component $d = 1, \dots, D$, where

$$\mathbf{C}_{\text{QMC}}(N, \alpha) := 2\sqrt{D}K \left(\mathcal{D}_N^* + 4(\mathcal{D}_N^*)^{1/D} \right) \frac{\alpha^{\frac{D}{\beta}} + 2K(1 + 2K)\|\nabla f\|_\infty \alpha^{1+\frac{D}{\beta}}}{C_0(\alpha)}. \quad (3.35)$$

Assume in addition that the assumptions of Lemma 3.7 are satisfied. Then

$$|(F_{\infty,\alpha}(z, s))_d| \leq \mathbf{C}_{\text{Lap}}(\alpha),$$

where the constant $\mathbf{C}_{\text{Lap}}(\alpha)$ is defined by

$$\mathbf{C}_{\text{Lap}}(\alpha) := \frac{\alpha^{-\frac{D+\gamma}{\beta}}}{C_0(\alpha)} C_{f,D,\beta,\gamma} + \frac{\alpha^{\frac{1}{\beta}}}{C_0(\alpha) \exp(\alpha c)} (2K)^{D+1}. \quad (3.36)$$

Here, the quantity $C_{f,D,\beta,\gamma}$ appearing in the definition of $\mathbf{C}_{\text{Lap}}(\alpha)$ is given explicitly by

$$C_{f,D,\beta,\gamma} := C_f \frac{2}{\beta} \pi^{\frac{D}{2}} \left(\frac{m_f}{2} \right)^{-\frac{D+1+\beta+\gamma}{\beta}} \frac{\Gamma((D+1+\beta+\gamma)/\beta)}{\Gamma(D/2)} \quad (3.37)$$

and depends only on structural properties of the objective function f and the dimension D .

Combining this bound with the quasi-Monte Carlo estimate derived previously and again using the fact that the exponential function is bounded by one for non-positive arguments, we obtain the component-wise estimate

$$|(F_{N,\alpha}(\bar{x}_s^N, s))_d| \leq \mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha). \quad (3.38)$$

Integrating this estimate over time yields

$$\int_0^t e^{\lambda s} |(F_{N,\alpha}(\bar{x}_s^N, s))_d| ds \leq \frac{\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)}{\lambda} (e^{\lambda t} - 1).$$

Substituting this estimate into equation (3.34) allows us to bound each component of the deviation from the minimizer from above by

$$\begin{aligned} (\bar{x}_t^N - x^*)_d &\leq e^{-\lambda t} (\bar{x}_0^N - x^*)_d + \lambda e^{-\lambda t} \int_0^t e^{\lambda s} |(F_{N,\alpha}(\bar{x}_s^N, s))_d| ds, \\ &\leq e^{-\lambda t} (\bar{x}_0^N - x^*)_d + (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda t}) \end{aligned} \quad (3.39)$$

and, analogously, from below by

$$(\bar{x}_t^N - x^*)_d \geq e^{-\lambda t} (\bar{x}_0^N - x^*)_d - (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda t}). \quad (3.40)$$

The above bounds form the basis for a quantitative convergence analysis of the average particle position $\bar{x}_s^N$ of the QMC²BO dynamics towards the minimizer x^* . In particular, for a fixed time horizon $[0, T]$, they allow us to verify the existence of a radius $0 < \delta \leq \delta_{\text{loc}}$ such that

$$B_\delta^\infty(x^*) \subseteq \bar{B}_{Ke^{-\lambda s}}^\infty(\bar{x}_s^N),$$

holds for all $s \in [0, T]$, which is a necessary condition required for the application of Lemma 3.7.

Before stating and proving the final result in Theorem 3.7, we briefly summarize the overall setting underlying the QMC²BO dynamics.

Method 3.6. [QMC²BO: Quasi-Monte Carlo Consensus-Based Optimization] Let $N = 2^m$, $m \in \mathbb{N}$, denote the number of particles in dimension D and let $[0, T]$ be a fixed time horizon. The QMC²BO method is defined as a deterministic interacting particle system $x_t^1, \dots, x_t^N$, $t \in [0, T]$, evolving according to

$$\frac{d}{dt}x_t^n = -\lambda(x_t^n - \nu_f^\alpha(x_t)), \quad n = 1, \dots, N,$$

where $\lambda, \alpha > 0$ are fixed parameters. The consensus point $\nu_f^\alpha(x_t)$ is defined by (1.3). The initial particle positions are given by $x_0^n = \mathfrak{s}(\mathbf{p}^n)$, where the point sequence $\mathcal{S} = (\mathbf{p}^n)_{n \in \mathbb{N}} \subset [0, 1]^D$ is uniformly distributed modulo one and the linear mapping $\mathfrak{s}$, defined in (1.2), maps the unit cube $[0, 1]^D$ onto the bounded domain $[-K, K]^D$ for some $K \in \mathbb{R}$.

For the QMC²BO method described in Method 3.6, we now establish a quantitative convergence result for the average particle position. Under additional regularity assumptions on the objective function f and for sufficiently large values of the concentration parameter α , we show that the average particle converges exponentially fast to the global minimizer of f , with an explicit error bound depending on N and α .

Theorem 3.7. Consider the QMC²BO method described in Methode 3.6. Let $f : \mathbb{R}^D \rightarrow \mathbb{R}$ be locally Lipschitz continuous, differentiable and satisfy Assumptions 3.4 with radius

$$\delta = \min \left(\delta_{\text{loc}}, e^{-\lambda T} \left(\frac{K(1-\rho)}{2} - \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| \right) \right).$$

Furthermore, let α^* be such that

$$(e^{\lambda T} - 1)(\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) \leq \frac{K(1-\rho)}{2}$$

holds for all $\alpha \geq \alpha^*$. Here, the constants $\mathbf{C}_{\text{QMC}}(N, \alpha)$ and $\mathbf{C}_{\text{Lap}}(\alpha)$ are defined in (3.35) and (3.36), respectively.

Then, for any $t \in [0, T]$, the average particle position satisfies

$$\lim_{N, \alpha \rightarrow \infty} \bar{x}_t^N = (1 - e^{-\lambda t})x^* + e^{-\lambda t}\bar{x}_0^\infty$$

with $\lim_{N \rightarrow \infty} \bar{x}_0^N = \bar{x}_0^\infty$. More precisely, the deviation of the average particle from the minimizer admits the bound

$$\|\bar{x}_t^N - x^*\| \leq e^{-\lambda t} \|\bar{x}_0^N - x^*\| + \sqrt{D}(\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha))(1 - e^{-\lambda t}),$$

which explicitly quantifies the contributions of the quasi-Monte Carlo approximation and the Laplace method error. In other words, the average particle $\bar{x}_t^N$ of the QMC²BO dynamics converges exponentially fast with rate λ to the global minimizer x^* of f .

Proof. The main idea of the proof is to apply Lemma 3.3 and Lemma 3.7 with the choice $z = \bar{x}_s^N$. Since the initial particle positions $x_0^1, \dots, x_0^N$ lie in the bounded domain $[-K, K]^D$, their average $\bar{x}_0^N$ belongs to $[-K, K]^D$ as well. Moreover, the convex hull of the particle system contracts over time, implying that $\bar{x}_s^N \in [-K, K]^D$ for all $s \in [0, T]$. As a consequence, the assumptions of Lemma 3.3 are satisfied.

In order to also apply Lemma 3.7, we must verify that our choice of δ and any $\alpha \geq \alpha^*$ ensures the cube inclusion

$$B_\delta^\infty(x^*) \subseteq \bar{B}_{Ke^{-\lambda s}}^\infty(\bar{x}_s^N)$$

for any $s \in [0, T]$. Geometrically, this condition is equivalent to requiring that the cube centered at x^* with side length 2δ is contained in the cube centered at $\bar{x}_s^N$ with side length $2Ke^{-\lambda s}$. By symmetry of the cube, this inclusion holds if and only if

$$x^* + \delta \mathbf{1}_D \leq \bar{x}_s^N + Ke^{-\lambda s} \mathbf{1}_D, \quad \bar{x}_s^N - Ke^{-\lambda s} \mathbf{1}_D \leq x^* - \delta \mathbf{1}_D.$$

By the assumptions on α^* and δ , we obtain

$$\begin{aligned} K(1 - \rho) &= \frac{K(1 - \rho)}{2} + \frac{K(1 - \rho)}{2} \\ &\geq e^{\lambda T} \delta + \max_{d=1, \dots, D} |(\bar{x}_0^N)_d| + (e^{\lambda T} - 1) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) \\ &\geq e^{\lambda s} \delta + |(\bar{x}_0^N)_d| + (e^{\lambda s} - 1) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)), \end{aligned}$$

where, in the last step, we have used that all involved quantities are positive and that $s \in [0, T]$. Moreover, by Assumption (A3), the minimizer satisfies $x^* \leq K_\rho \mathbf{1}_D$, where $K_\rho = K\rho$. Combining this with the previous inequality yields the component-wise bound

$$K - (x^*)_d \geq K(1 - \rho) \geq e^{\lambda s} \delta - (\bar{x}_0^N)_d + (e^{\lambda s} - 1) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha))$$

for each $d = 1, \dots, D$. Rearranging the terms and multiplying by $e^{-\lambda s}$ gives

$$K e^{-\lambda s} - \delta \geq (x^* - \bar{x}_0^N)_d e^{-\lambda s} + (1 - e^{-\lambda s}) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)).$$

Furthermore, using the lower component-wise estimate (3.40), we obtain

$$K e^{-\lambda s} - \delta \geq (x^* - \bar{x}_s^N)_d,$$

which, after rearranging, is exactly the first component-wise condition

$$(x^*)_d + \delta \leq (\bar{x}_s^N)_d + K e^{-\lambda s}.$$

required for the cube inclusion

Analogously, Assumption (A3) implies $x^* \geq -K_\rho \mathbf{1}_D$, which leads to the component-wise inequality

$$K + (x^*)_d \geq K(1 - \rho) \geq e^{\lambda s} \delta + (\bar{x}_0^N)_d + (e^{\lambda s} - 1) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha))$$

for $d = 1, \dots, D$. Rearranging, multiplying by $e^{-\lambda s}$, and applying the upper bound (3.39) yields

$$K e^{-\lambda s} - \delta \geq (\bar{x}_0^N - x^*)_d e^{-\lambda s} + (1 - e^{-\lambda s}) (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)) \geq (\bar{x}_s^N - x^*)_d,$$

which, after a final rearrangement, coincides with the second component-wise condition

$$(\bar{x}_s^N)_d - K e^{-\lambda s} \leq (x^*)_d - \delta$$

required for the validity of (3.3).

Finally, all assumptions are verified and we may apply both Lemma 3.3 and Lemma 3.7. Starting from the representation (3.6), we obtain

$$\begin{aligned} \lim_{N, \alpha \rightarrow \infty} \bar{x}_t^N &= (1 - e^{-\lambda t}) x^* + e^{-\lambda t} \lim_{N \rightarrow \infty} \bar{x}_0^N + e^{-\lambda t} \int_0^t e^{\lambda s} \lim_{N, \alpha \rightarrow \infty} F_{N, \alpha}(\bar{x}_s^N, s) ds. \\ &= (1 - e^{-\lambda t}) x^* + e^{-\lambda t} \bar{x}_0^\infty + e^{-\lambda t} \int_0^t e^{\lambda s} \lim_{\alpha \rightarrow \infty} F_{\infty, \alpha}(\bar{x}_s^N, s) ds. \\ &= (1 - e^{-\lambda t}) x^* + e^{-\lambda t} \bar{x}_0^\infty, \end{aligned}$$

where we have used the identity (3.24) and applied Lemma 3.7 with the choice $z = \bar{x}_s^N$.

To derive the explicit convergence rate, we again consider the representation (3.6) and apply the triangle inequality, which yields

$$\begin{aligned} \|\bar{x}_t^N - x^*\| &= \left\| e^{-\lambda t} (\bar{x}_0^N - x^*) + e^{-\lambda t} \int_0^t e^{\lambda s} F_{N, \alpha}(\bar{x}_s^N, s) ds \right\| \\ &\leq e^{-\lambda t} \|\bar{x}_0^N - x^*\| + e^{-\lambda t} \int_0^t e^{\lambda s} \|F_{N, \alpha}(\bar{x}_s^N, s)\| ds. \end{aligned}$$

Using the definition of the Euclidean norm together with the component-wise estimate (3.38), we further obtain

$$\|F_{N, \alpha}(\bar{x}_s^N, s)\| = \sqrt{\sum_{d=1}^D |(F_{N, \alpha}(\bar{x}_s^N, s))_d|^2} \leq \sqrt{D} (\mathbf{C}_{\text{Lap}}(\alpha) + \mathbf{C}_{\text{QMC}}(N, \alpha)).$$

Integrating this bound over time leads to

$$\int_0^t e^{\lambda s} \|F_{N,\alpha}(\bar{x}_s^N, s)\| ds \leq \sqrt{D} (C_{\text{Lap}}(\alpha) + C_{\text{QMC}}(N, \alpha)) (e^{\lambda t} - 1).$$

Combining the above estimates, we finally arrive at

$$\|\bar{x}_t^N - x^\star\| \leq e^{-\lambda t} \|\bar{x}_0^N - x^\star\| + \sqrt{D} (C_{\text{Lap}}(\alpha) + C_{\text{QMC}}(N, \alpha)) (1 - e^{-\lambda t}),$$

which establishes the claimed exponential convergence of the average particle position towards the minimizer, with an explicit rate and constants depending on N and α . $\square$

4 Numerical Experiments for the QMC²BO Method

5 Discussion and Further Results from the Literature

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